University of Waterloo

ACTSC 446

Mathematics of Financial Markets

Lecture Notes

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Standing conventions

Unless stated otherwise, $0 \le t \le T$, $S_u \ge 0$, $K \ge 0$, and r is a constant continuously compounded risk-free rate. When dividends are continuous, $\delta \ge 0$ denotes a constant proportional dividend yield. The stock price is *ex-dividend*. We explicitly impose $r \ge 0$ whenever a proof needs it. Discounted European bounds and forward formulas themselves allow negative r.

What is expanded or corrected

The bank account $B_u = e^{ru}$ is distinguished from a zero-coupon bond paying one dollar at maturity. Arbitrage includes nonnegative payoffs with a strictly positive probability of gain, and strategies must be self-financing. Discrete-dividend forward formulas require dividends known at the valuation time. Forward delivery prices are distinguished from contract values. American exercise arguments cover the short writer's obligations at every permitted exercise time. Payoff, premium, and financed profit are kept separate. The final sections clarify the lecture's butterfly convention and the limitations of claims about volatility and risk-adjusted performance.

Notation

Symbol	Meaning
S_u	Ex-dividend stock price at time u
$B_u = e^{ru}$	Bank account starting at one dollar at time zero
$D(t,T) = e^{-r(T-t)}$	Price at t of one dollar delivered at T
$A(t,T)$	Price at t of a prepaid claim to one share at T
$F(t,T)$	Delivery price of a new zero-value forward
$f_t(K,T)$	Value at t of a long forward with fixed delivery price K
$c(K,t,T),\ p(K,t,T)$	European call and put prices
$C(K,t,T),\ P(K,t,T)$	American call and put prices
$x^+ = \max\{x, 0\}$	Positive part of a real number

Chapter 1

Introduction to Derivatives Markets

1.1 Financial markets and trading conventions

1.1.1 Assets, time, and uncertainty

Definition 1.1 (Asset and price process)**.** An asset is a financial instrument that can be bought or sold. Its price process $(S_u)_{u\geq 0}$ specifies its market value through time. At a future time, S_u may be random. At the valuation time t, the observed S_t is known to the trader.

One may formalize uncertainty on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_u)_{u\geq 0}, \mathbb{P})$, with S_u measurable with respect to $\mathcal{F}_u$. This says that the price at u is observable using information available by u. Part I does not specify the law of (S_u); most arguments below hold path by path. At a later valuation time t, probabilistic statements are understood relative to the information then available. In particular, a claim of strict price dominance requires a positive conditional probability of strict payoff dominance at that valuation state. At time zero this reduces to the usual unconditional statement.

Definition 1.2 (Model-free argument)**.** A pricing argument is model-free here if it relies on trading assumptions and payoff identities or inequalities, without specifying a probability distribution or stochastic differential equation for the underlying.

Model-free does not mean assumption-free. Dividend rules, financing opportunities, short-sale access, settlement conventions, and exercise rights all matter.

Definition 1.3 (Perfect capital market conventions)**.** For the arguments in these notes, assume:

(i) no transaction costs and a common buying and selling price;

(ii) arbitrary fractional positions and unrestricted short positions;

(iii) execution at observed prices, with no delay or price impact;

(iv) borrowing and lending at the same specified risk-free rate;

(v) no counterparty default on promised contractual cash flows;

(vi) no trading advantage arising from different information sets.

Continuous-time exercise arguments additionally require the ability to transact when exercise occurs.

Remark 1.1 (Credit risk versus price risk). Removing counterparty default risk does not make a stock risk-free. Its market value may still fluctuate or become zero. A stock price loss is different from failure to honour a promised derivative payment.

1.1.2 The bank account and zero-coupon bonds

Definition 1.4 (Bank account)**.** The risk-free bank account with $B_0 = 1$ is

$$B_u = e^{ru}.$$

An amount a invested at time t has value $aB_u/B_t = ae^{r(u-t)}$ at $u \geq t$. A negative amount represents borrowing.

Definition 1.5 (Zero-coupon bond)**.** A zero-coupon bond with maturity T and face value one pays exactly one dollar at T, with no intermediate payments. Its time-t price is denoted by $D(t, T)$.

Proposition 1.1 (Discounting a deterministic payment)**.** *In an arbitrage-free market with the stated bank account,*

$$D(t, T) = \frac{B_t}{B_T} = e^{-r(T-t)}.$$

A deterministic payment a at T therefore has value $ae^{-r(T-t)}$ at t.

Proof. Purchase $1/B_T$ units of the bank account. The cost and terminal value are

$$\text{cost at } t = \frac{B_t}{B_T} = e^{-r(T-t)},$$
$$\text{value at } T = \frac{B_T}{B_T} = 1.$$

A bond with the same cash flows must have the same price by the law of one price, proved in Section 1.2. Multiplying the position by a gives the general formula. □

Remark 1.2 (Two different assets). The lecture writes $B_t = e^{rt}$ and calls this a risk-free bond. We use the more precise name *bank account*. A bond that pays one at T has price $e^{-r(T-t)}$, whereas one dollar placed in the bank account at zero becomes e^{rT} at T.

Remark 1.3 (The sign of the interest rate). For $r > 0$, a future dollar costs less than a current dollar. For $r = 0$, the two have equal value. For $r < 0$, the future dollar costs more. None of these possibilities contradicts no-arbitrage by itself.

1.1.3 Portfolios, cash flows, and self-financing

Definition 1.6 (Portfolio value)**.** For assets with prices $S_u^1, \ldots, S_u^m$ and holdings $\theta_u^1, \ldots, \theta_u^m$, the marked value is

$$V_u = \sum_{i=1}^{m} \theta_u^i S_u^i.$$

Positive holdings are long positions; negative holdings are short positions.

For example, three shares and 200 units of the bank account have value $3S_u + 200B_u$. Three shares financed by borrowing 200 dollars at time zero have value $3S_u - 200B_u$. At a later starting time t, a loan of 200 dollars instead contributes $-200B_u/B_t$.

Definition 1.7 (Self-financing strategy)**.** A strategy is self-financing if, after its initial setup, changes in holdings and contractual obligations are funded entirely by asset sales, borrowing, and cash flows from its existing positions. No outside money is injected or withdrawn without being recorded.

At a rebalancing time u with no simultaneous dividend, changing holdings from (θ, β) to (θ', β') requires

$$\begin{aligned}(\theta' - \theta)S_u + (\beta' - \beta)B_u &= 0,\\ \theta' S_u + \beta' B_u &= \theta S_u + \beta B_u.\end{aligned}$$

Thus rebalancing redistributes wealth; it does not create wealth. If dividends are received, they can finance additional shares or be put in the bank account.

Remark 1.4 (Value versus cash flow). Buying an asset priced at v produces an initial cash flow of $-v$ but acquires value v. Selling it produces cash flow $+v$ and a position with value $-v$. Cash-flow tables below use *positive for cash received*. A portfolio's initial cost has the opposite sign to its setup cash flow.

Remark 1.5 (Short sales and dividends). A short seller owes any distributions paid to the owner of the borrowed stock. A short share contributes $-S_T$ upon closing at T, but this is not its entire cash-flow history if dividends occur before T.

1.1.4 Derivatives as contractual claims

Definition 1.8 (Derivative)**.** A derivative is a contract whose payments are determined by specified underlying market quantities. A European terminal claim may be written as

$$H = h(S_T),$$

while a path-dependent claim may depend on the whole observed path $(S_u)_{0 \le u \le T}$.

Underlying quantities may include stocks, bonds, currencies, indices, commodities, or rates. Derivatives can transfer risk, create compensation incentives, or take speculative exposures. A small initial premium need not mean a small exposure: contractual payments can be much larger than the setup cost.

1.2 No-arbitrage and the comparison principle

Source coverage: lecture slides 8–12..

1.2.1 A precise definition of arbitrage

Definition 1.9 (Arbitrage)**.** Starting at time t, an arbitrage is an admissible self-financing strategy with zero initial wealth, terminal wealth $Z_T \geq 0$ almost surely, and

$$\mathbb{P}(Z_T > 0) > 0.$$

Admissibility rules out strategies such as unlimited doubling schemes. All explicit constructions here use finite positions and financed contractual obligations.

A strategy with negative initial cost and nonnegative future cash flows also generates arbitrage: invest the initial receipt. If its cost is $-a < 0$, the additional bank holding pays $ae^{r(T-t)} > 0$ at T.

Remark 1.6 (Relation to the lecture definition)*.* The slides use $V_0 \leq 0$ and $V_T > 0$. Guaranteed positive terminal wealth is a particularly transparent arbitrage. The standard probabilistic definition also includes a payoff that is positive only on an event of positive probability and never negative. Our explicit mispricing constructions usually produce a deterministic positive bank balance in every state.

1.2.2 The law of one price

Theorem 1.1 (Law of one price)**.** *Suppose two attainable strategies X and Y have identical remaining cash flows, including identical terminal liquidation values, and no unaccounted intermediate payments. Then their prices at time t are equal.*

Proof. Suppose $X_t > Y_t$ and let $a = X_t - Y_t > 0$. Buy Y, sell X, and invest a in the bank account. The resulting strategy has value

$$Z_u = Y_u - X_u + a\frac{B_u}{B_t}.$$

At setup,

$$\begin{aligned} Z_t &= Y_t - X_t + (X_t - Y_t) \\ &= 0. \end{aligned}$$

Because the positions have matching cash flows, they cancel, leaving

$$\begin{aligned} Z_T &= Y_T - X_T + a\frac{B_T}{B_t} \\ &= ae^{r(T-t)} > 0. \end{aligned}$$

This is arbitrage. If $Y_t > X_t$, reverse the positions. Therefore $X_t = Y_t$. □

Remark 1.7 (Why terminal values alone may not suffice)*.* Two assets with equal ex-dividend values at T may have paid different dividends before T. Their terminal prices alone do not identify their total cash flows. Either match all intermediate payments or explicitly reinvest them and compare terminal wealth.

1.2.3 Payoff dominance and strict inequalities

Proposition 1.2 (Monotonicity of attainable prices)**.** *Let X and Y be terminal-payoff claims with no intermediate payments. If $X_T \geq Y_T$ almost surely, then $X_t \geq Y_t$. If additionally $\mathbb{P}(X_T > Y_T) > 0$, then $X_t > Y_t$ under the stated arbitrage definition.*

Proof. If $X_t < Y_t$, buy X, sell Y, and invest the initial receipt $Y_t - X_t$. Then

$$\begin{aligned} Z_t &= X_t - Y_t + (Y_t - X_t) = 0, \\ Z_T &= X_T - Y_T + (Y_t - X_t)e^{r(T-t)} > 0. \end{aligned}$$

Thus $X_t \geq Y_t$. If the terminal inequality is strict with positive probability but $X_t = Y_t$, the zero-cost position $X - Y$ already has nonnegative payoff and positive probability of gain. This is also arbitrage, proving strictness. □

Corollary 1.1 (Nonnegative payoff and deterministic bounds)**.** *If $H \geq 0$, its price is nonnegative. More generally, if deterministic constants satisfy $a \leq H \leq b$, then*

$$aD(t,T) \leq \text{price}_t(H) \leq bD(t,T).$$

Proof. Compare H with zero, or with bonds paying a and b at T. □

Remark 1.8 (Weak versus strict wording). A payoff that is merely "higher" in the sense $X_T \geq Y_T$ supports a weak price inequality. Strictness needs a genuine strictly positive gain on an event that is not null. This distinction matters when payoffs agree on every reachable stock-price state.

1.2.4 Uniqueness of the risk-free rate

Proposition 1.3 (A common risk-free rate)**.** *Two freely accessible risk-free accounts, with identical currency, maturity, and borrowing/lending permissions, cannot offer different continuously compounded rates.*

Proof. Suppose $r_2 > r_1$. Borrow one dollar at r_1 and lend it at r_2. The initial cash flow is $1 - 1 = 0$, and the terminal net wealth is

$$\begin{aligned} Z_T &= e^{r_2(T-t)} - e^{r_1(T-t)} \\ &= e^{r_1(T-t)} \left(e^{(r_2 - r_1)(T-t)} - 1 \right) > 0. \end{aligned}$$

This contradicts no-arbitrage. Interchanging the accounts rules out $r_1 > r_2$. □

Remark 1.9. Different borrowing and lending rates in an imperfect market do not automatically imply arbitrage. The proof requires borrowing at the lower rate and lending at the higher rate in unrestricted amounts.

1.2.5 A reusable proof method

To prove a price identity, first match cash flows. To prove a bound, construct a portfolio that covers the claim in every state. If the claimed price relation fails, buy the cheaper cash-flow stream, sell the more expensive one, and invest the difference. For American options, "every state" must include *every permitted exercise time of the short position.*

1.3 Forwards, dividends, and futures

Source coverage: lecture slides 14–40..

1.3.1 The forward contract and its two prices

Definition 1.10 (Forward contract)**.** A forward with delivery date T and delivery price K obliges its long holder to pay K and receive one unit of the underlying at T. The short holder must deliver that unit and receives K.

The net values exchanged at maturity are

$$\begin{aligned} H_T^{\text{long}} &= S_T - K, \\ H_T^{\text{short}} &= K - S_T = -H_T^{\text{long}}. \end{aligned}$$

Physical delivery and cash settlement have the same net value in the frictionless setting.

Definition 1.11 (Forward price and forward value)**.** The *forward price* $F(t,T)$ is the delivery price that makes a new forward have zero value at time t. The *forward value* $f_t(K,T)$ is the market value at t of an existing contract whose delivery price is the fixed number K.

Thus $f_t(F(t,T),T) = 0$. The number K does not reset when market conditions change; the value of the existing contract does.

1.3.2 No-dividend forward replication

Theorem 1.2 (Forward value without dividends)**.** *If the stock pays no dividends on* $(t,T]$*, then*

$$\begin{aligned} f_t(K,T) &= S_t - KD(t,T), \\ F(t,T) &= S_t e^{r(T-t)}. \end{aligned}$$

Proof. Buy one share and borrow $KD(t,T)$. The loan grows to K at maturity. The portfolio has cost $S_t - KD(t,T)$ and payoff

$$\begin{aligned} V_T &= S_T - KD(t,T)e^{r(T-t)} \\ &= S_T - K. \end{aligned}$$

This replicates the long forward, so the law of one price gives the first formula. A new forward has value zero, hence

$$\begin{aligned} 0 &= S_t - F(t,T)e^{-r(T-t)}, \\ F(t,T)e^{-r(T-t)} &= S_t, \\ F(t,T) &= S_t e^{r(T-t)}. \end{aligned}$$

□

Position	Cash flow at t	Cash flow at T
Buy one share	$-S_t$	S_T
Borrow $KD(t,T)$	$+KD(t,T)$	$-K$
Total	$-S_t + KD(t,T)$	$S_T - K$

Arbitrage when the quoted delivery price is too high

Suppose a newly offered zero-cost forward has delivery price $K > S_t e^{r(T-t)}$. Buy one share, borrow S_t, and short the forward. Then

$$\begin{aligned}\text{initial cash flow} &= -S_t + S_t + 0 = 0,\\ \text{terminal wealth} &= S_T - S_t e^{r(T-t)} + (K - S_T)\\ &= K - S_t e^{r(T-t)} > 0.\end{aligned}$$

Arbitrage when the quoted delivery price is too low

If $K < S_t e^{r(T-t)}$, short one share, invest S_t, and go long the forward. Then

$$\begin{aligned}\text{initial cash flow} &= S_t - S_t + 0 = 0,\\ \text{terminal wealth} &= -S_T + S_t e^{r(T-t)} + (S_T - K)\\ &= S_t e^{r(T-t)} - K > 0.\end{aligned}$$

Both directions are necessary: one construction gives one inequality, and its reverse gives the other.

Example 1.1 (A contract need not remain worth zero)**.** Let $S_0 = 100$, $r = 0.05$, and $T = 1$. The original delivery price is $K = 100e^{0.05} \approx 105.1271$. At $t = 0.5$, suppose $S_t = 110$. Then

$$\begin{aligned}f_{0.5}(K, 1) &= 110 - 105.1271e^{-0.025}\\ &\approx 7.4685.\end{aligned}$$

The original long forward is now valuable. Its zero initial cost was a statement about inception, not all later times.

1.3.3 Continuous proportional dividends

Definition 1.12 (Continuous dividend yield)**.** A stock with constant dividend yield δ pays $\delta S_u \,\mathrm{d}u$ per share during a short interval of length $\mathrm{d}u$. The quoted S_u excludes cash already paid as dividends.

Lemma 1.1 (Reinvestment grows the share count)**.** *If dividends are immediately reinvested, a position of n_t shares grows to*

$$n_u = n_t e^{\delta(u-t)}.$$

In particular, buying $e^{-\delta(T-t)}$ shares at t and reinvesting dividends produces exactly one share at T.

Proof. The dividend received on n_u shares is $n_u \delta S_u \,\mathrm{d}u$. Reinvesting it buys

$$\begin{aligned}\mathrm{d}n_u &= \frac{n_u \delta S_u \,\mathrm{d}u}{S_u}\\ &= \delta n_u \,\mathrm{d}u.\end{aligned}$$

For a positive share price, solving this ordinary differential equation gives

$$\begin{aligned}\frac{\mathrm{d}n_u}{n_u} &= \delta\,\mathrm{d}u,\\ \log n_u - \log n_t &= \delta(u-t),\\ n_u &= n_t e^{\delta(u-t)}.\end{aligned}$$

Choose $n_t = e^{-\delta(T-t)}$ to obtain $n_T = 1$. The continuous-yield model convention extends this construction through zero-price states; there is then no positive dividend cash flow to finance. Equivalently, the reinvested position's wealth and dividend-financed share rule specify the strategy directly. □

Definition 1.13 (Prepaid forward)**.** A prepaid forward is a claim paid for at t that delivers one share at T, with no delivery payment at T and no intervening dividends to its holder. Denote its price by $A(t,T)$.

The lemma and replication give

$$A(t,T) = S_t e^{-\delta(T-t)}.$$

This is not generally S_t: the current shareholder has dividend rights that the prepaid-forward holder does not receive.

Theorem 1.3 (Forward pricing with continuous dividends)**.** *Under the continuous-yield convention,*

$$\begin{aligned}f_t(K,T) &= S_t e^{-\delta(T-t)} - Ke^{-r(T-t)},\\ F(t,T) &= S_t e^{(r-\delta)(T-t)}.\end{aligned}$$

Proof. Buy $e^{-\delta(T-t)}$ shares and reinvest dividends until T. Borrow $Ke^{-r(T-t)}$. The stock holding becomes one share and the loan becomes K, so the terminal payoff is $S_T - K$. Consequently,

$$\begin{aligned}f_t(K,T) &= A(t,T) - KD(t,T)\\ &= S_t e^{-\delta(T-t)} - Ke^{-r(T-t)}.\end{aligned}$$

Set $K = F(t,T)$ and solve $f_t(K,T) = 0$:

$$\begin{aligned}F(t,T)e^{-r(T-t)} &= S_t e^{-\delta(T-t)},\\ F(t,T) &= S_t e^{r(T-t)-\delta(T-t)}\\ &= S_t e^{(r-\delta)(T-t)}.\end{aligned}$$

□

Remark 1.10 (Why dividends reduce the delivery price). Holding a share delivers dividends; holding a forward does not. Financing the share costs interest, while its dividends offset part of that cost. The net carry rate is $r - \delta$. The forward price is below spot when $\delta > r$, equal to spot when $\delta = r$, and above spot when $\delta < r$.

Example 1.2. For $S_t = 100$, $T - t = 0.5$, $r = 0.04$, and $\delta = 0.015$,

$$\begin{aligned}A(t,T) &= 100e^{-0.015(0.5)} \approx 99.2528,\\ F(t,T) &= 100e^{(0.04-0.015)(0.5)} \approx 101.2578.\end{aligned}$$

The prepaid price and the delivery price refer to different payment dates.

1.3.4 Known discrete cash dividends

Theorem 1.4 (Forward pricing with deterministic cash dividends)**.** *Suppose the stock pays known cash amounts d_i at dates $t < t_1 < \cdots < t_m \leq T$. Assume the forward holder does not receive these dividends and that a dividend at T is paid before delivery. Write*

$$I_t = \sum_{i=1}^{m} d_i e^{-r(t_i - t)}.$$

Then

$$\begin{aligned} A(t,T) &= S_t - I_t, \\ f_t(K,T) &= S_t - I_t - Ke^{-r(T-t)}, \\ F(t,T) &= (S_t - I_t)e^{r(T-t)} \\ &= S_t e^{r(T-t)} - \sum_{i=1}^{m} d_i e^{r(T-t_i)}. \end{aligned}$$

Proof. Buy one share and borrow $d_i e^{-r(t_i-t)}$ separately for each dividend date t_i. At t_i, the corresponding debt is

$$d_i e^{-r(t_i-t)} e^{r(t_i-t)} = d_i,$$

which is repaid using that dividend. There are no net intermediate cash flows and one share remains at T. The initial cost is $S_t - I_t$, proving the prepaid-forward formula. Borrowing an additional $Ke^{-r(T-t)}$ changes the terminal payoff to $S_T - K$, yielding the forward-value formula. Finally,

$$\begin{aligned} F(t,T) &= \left(S_t - \sum_{i=1}^{m} d_i e^{-r(t_i-t)} \right) e^{r(T-t)} \\ &= S_t e^{r(T-t)} - \sum_{i=1}^{m} d_i e^{r[(T-t)-(t_i-t)]} \\ &= S_t e^{r(T-t)} - \sum_{i=1}^{m} d_i e^{r(T-t_i)}. \end{aligned}$$

□

Alternative cash-and-carry derivation

Borrow S_t, buy one share, reinvest every cash dividend in the bank, and short a zero-cost forward with delivery price K. Terminal wealth is

$$\begin{aligned} Z_T &= S_T + \sum_i d_i e^{r(T-t_i)} - S_t e^{r(T-t)} + (K - S_T) \\ &= K - S_t e^{r(T-t)} + \sum_i d_i e^{r(T-t_i)}. \end{aligned}$$

This quantity is deterministic. If it is positive, use the strategy; if negative, reverse it. No-arbitrage therefore forces it to equal zero.

Example 1.3. Let $S_0 = 100$, $r = 0.05$, $T = 1$, and let the only dividend be two dollars at $t_1 = 0.5$. Then

$$\begin{aligned} F(0,1) &= 100e^{0.05} - 2e^{0.05(1-0.5)} \\ &\approx 103.0765. \end{aligned}$$

Subtracting merely two dollars from the no-dividend forward price would omit the interest earned on the dividend between 0.5 and 1.

Remark 1.11 (Uncertain dividends). If the dividend amounts are random and not otherwise hedged, the displayed terminal sum is random. No-arbitrage does not let us set it equal to a fixed delivery price state by state. Replacing those dividends by physical expectations is not a model-free pricing argument. Additional tradable claims or modelling assumptions are needed.

1.3.5 Futures, marking to market, and margin

Definition 1.14 (Futures settlement)**.** A futures contract is standardized and exchange-traded, with gains and losses settled through marking to market. If futures quotes at settlement dates $t_0 < \cdots < t_n = T$ are $G_0, \ldots, G_n$, a unit long position receives

$$G_i - G_{i-1}$$

at date t_i. For a cash-settled contract on the terminal spot value, $G_n = S_T$.

The undiscounted sum telescopes:

$$\begin{aligned} \sum_{i=1}^{n} (G_i - G_{i-1}) &= G_n - G_0 \\ &= S_T - G_0. \end{aligned}$$

But if each settlement is financed or invested until T, total terminal wealth is instead

$$\sum_{i=1}^{n} (G_i - G_{i-1}) e^{r(T-t_i)}.$$

These are different expressions. A forward makes one payment $S_T - K$ at T; a futures position produces a stream of payments before T.

Example 1.4 (Settlement timing matters)**.** Suppose $G_0 = 100$, $G_1 = 110$ at $t_1 = 0.5$, $G_2 = 105$ at $T = 1$, and $r = 0.05$. Then the long futures position receives 10 and later pays 5. Its accumulated settlement value is

$$10e^{0.05(0.5)} - 5 \approx 5.2532,$$

whereas a forward entered at $K = 100$ pays $105 - 100 = 5$. This comparison isolates settlement timing; it does not assert that these particular starting quotes are fair in a specified model.

Proposition 1.4 (Why deterministic-rate futures and forward quotes agree)**.** *In an idealized settlement model with deterministic financing rates, unrestricted fractional futures holdings, and $G_n = S_T$, the initial futures quote equals the forward delivery price for the same terminal underlying and delivery date.*

Proof. During $(t_{i-1}, t_i]$, hold $q_i = e^{-r(T-t_i)}$ futures contracts. Changes to the number of contracts at settlement have zero contract value; all gains and losses are put in the bank account, allowing borrowing when needed. The terminal value of settlement i is

$$q_i(G_i - G_{i-1})e^{r(T-t_i)} = G_i - G_{i-1}.$$

Summing produces $S_T - G_0$ from zero initial wealth. This replicates a zero-value forward with delivery price G_0. Hence $G_0 = F(t_0, T)$. □

Remark 1.12. With random rates, the required future accumulation factor need not be known at t_{i-1}. The argument fails, and equality is not automatic. Margin funding frictions can also affect the practical comparison.

Definition 1.15 (Margin)**.** Initial margin is collateral required to support a futures position. Variation margin transfers settlement gains and losses. A margin call requests additional collateral or funding when the account falls below the required level.

Zero contract value at inception does not mean zero cash resources are required. Margin is distinct from an option premium or an asset purchase price. A hedge can lose cash before it delivers its intended economic offset, creating a funding requirement.

Remark 1.13 (Institutional descriptions). Forwards are commonly privately negotiated, while futures use standardized exchange and clearing arrangements. Clearing and collateral mitigate default exposure rather than making it logically impossible. Being over the counter does not mean being unregulated. These institutional remarks motivate the model; they are not used to prove the pricing identities.

1.4 Options: rights, payoffs, and profit

Source coverage: lecture slides 41–53, 55–57..

1.4.1 Calls, puts, and exercise styles

Definition 1.16 (Call and put)**.** A call gives its holder the right, but not the obligation, to buy one unit of the underlying for K. A put gives its holder the right, but not the obligation, to sell one unit for K. The number K is the strike price, and T is the expiry date.

Definition 1.17 (Exercise styles)**.** A European option can be exercised only at T. An American option can be exercised at any permitted time in $[t, T]$. A Bermudan option can be exercised on a specified set of dates, including its final expiry date.

The long holder chooses whether to exercise. The short writer must honour that choice. These names describe exercise rights, not geographical trading locations.

Definition 1.18 (Exercise time)**.** In a stochastic formulation, an American exercise rule is a stopping time τ with values in $[t, T]$: for every u, the event $\{\tau \le u\}$ belongs to $\mathcal{F}_u$. Thus the decision cannot use information that arrives only later.

No optimal-stopping pricing theorem is needed here. To prove an American upper bound, we construct a hedge that works for every permitted τ.

1.4.2 The positive part and elementary payoff identities

Lemma 1.2 (Positive-part identities)**.** *For every $x \in \mathbb{R}$,*

$$x^+ = \begin{cases} x, & x \ge 0, \\ 0, & x < 0, \end{cases}$$
$$x^+ - (-x)^+ = x,$$
$$x^+ + (-x)^+ = |x|.$$

Moreover, $x \mapsto x^+$ is increasing, 1-Lipschitz, and convex.

Proof. If $x \ge 0$, the two positive parts are x and zero; if $x < 0$, they are zero and $-x$. This proves the identities. For $x \le y$, check three cases:

$$\begin{aligned} y \le 0: &\quad y^+ - x^+ = 0, \\ x \le 0 \le y: &\quad y^+ - x^+ = y \le y - x, \\ 0 \le x \le y: &\quad y^+ - x^+ = y - x. \end{aligned}$$

Thus $0 \le y^+ - x^+ \le y - x$, proving monotonicity and the Lipschitz bound. For $\lambda \in [0, 1]$, both $x \le x^+$ and $y \le y^+$ imply

$$\lambda x + (1 - \lambda)y \le \lambda x^+ + (1 - \lambda)y^+.$$

The right side is nonnegative, so taking the maximum of zero and the left side preserves the inequality:

$$(\lambda x + (1 - \lambda)y)^+ \le \lambda x^+ + (1 - \lambda)y^+.$$

This is convexity. □

Proposition 1.5 (Vanilla terminal payoffs)**.** *For a stock price $s \geq 0$ at exercise, the long call and long put payoffs are*

$$h_c(s) = (s-K)^+ = \begin{cases} 0, & s \leq K, \\ s-K, & s > K, \end{cases}$$

$$h_p(s) = (K-s)^+ = \begin{cases} K-s, & s < K, \\ 0, & s \geq K. \end{cases}$$

Short positions have payoffs $-h_c(s)$ and $-h_p(s)$.

Proof. A call holder can buy for K and acquire value s, producing net value $s-K$, or decline to exercise and receive zero. Choose the larger. A put holder can deliver value s for K, producing $K-s$, or receive zero. The writer pays the holder's contractual receipt. □

Remark 1.14 (A short call is not a long put)*.* A short call pays $-(s-K)^+$, whereas a long put pays $(K-s)^+$. Their values coincide only in some states. In particular, a short call has unbounded loss as $s \to \infty$, while a long put has nonnegative payoff bounded by K.

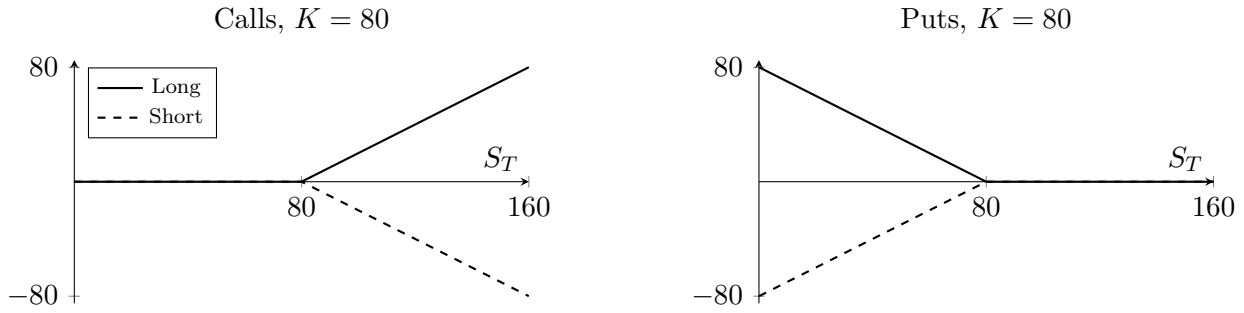


Figure 1.1: Terminal payoffs before accounting for the premium. Solid lines are long positions; dashed lines are short positions.

1.4.3 Moneyness, intrinsic value, and exercise

Definition 1.19 (Moneyness)**.** A call is in the money if $S_t > K$, at the money if $S_t = K$, and out of the money if $S_t < K$. A put is in the money if $S_t < K$, at the money if $S_t = K$, and out of the money if $S_t > K$.

The intrinsic values are $(S_t - K)^+$ and $(K - S_t)^+$. An out-of-the-money option has zero intrinsic value, not a negative optimal exercise payoff: the holder can decline an unfavourable exchange. European moneyness describes the current spot-strike comparison even though immediate exercise is unavailable.

Definition 1.20 (Option price notation)**.** Write $c_t = c(K,t,T)$ and $p_t = p(K,t,T)$ for European call and put prices, and $C_t = C(K,t,T)$ and $P_t = P(K,t,T)$ for American call and put prices.

Proposition 1.6 (More exercise rights cannot reduce value)**.** *For otherwise identical contracts,*

$$C_t \geq c_t, \qquad P_t \geq p_t.$$

If a Bermudan option includes T among its exercise dates, its value lies between the corresponding European and American values. Also,

$$C_t \geq (S_t - K)^+, \qquad P_t \geq (K - S_t)^+.$$

Proof. If the contract with more exercise rights were cheaper, buy it and sell the restricted contract. Exercise the long contract whenever the short contract is exercised, so the obligations cancel. Invest the initial receipt. This is arbitrage. For the intrinsic-value bounds, a holder can exercise immediately, or choose not to exercise and obtain at least zero. A price below either achievable nonnegative immediate receipt would give an immediate arbitrage. □

1.4.4 Payoff versus financed profit

Definition 1.21 (Terminal financed profit)**.** For a portfolio costing V_t at t, with terminal payoff H_T and no intermediate payments, its terminal profit relative to risk-free financing is

$$\Pi_T = H_T - V_t e^{r(T-t)}.$$

The corresponding discounted profit is $D(t,T)H_T - V_t$.

For long and short calls,

$$\Pi_T^{\text{long call}} = (S_T - K)^+ - c_t e^{r(T-t)},$$
$$\Pi_T^{\text{short call}} = c_t e^{r(T-t)} - (S_T - K)^+.$$

The long call can lose its entire premium plus its financing cost. The short call receives a premium initially but retains unbounded potential loss. For a put, replace $(S_T - K)^+$ by $(K - S_T)^+$.

Example 1.5 (The lecture's call-profit calculation)**.** Take $K = 260$ and premium $c_0 = 1.08$ per share. Ignoring financing, profit is

$$\begin{aligned}\Pi_T &= (S_T - 260)^+ - 1.08 \\ &= \begin{cases} -1.08, & S_T \leq 260, \\ S_T - 261.08, & S_T > 260. \end{cases}\end{aligned}$$

Thus the break-even price is 261.08. The simple premium return is

$$R = \frac{(S_T - 260)^+ - 1.08}{1.08}.$$

For $S_T \leq 260$, this is -1, or a loss of 100%. Including financing changes break-even to $260 + 1.08e^{rT}$. For a contract covering m shares, multiply all dollar amounts by m; break-even per share is unchanged.

Remark 1.15 (Reading an option chain). An option chain organizes quotes by strike and expiry, usually separating calls and puts. A bid is a price at which one can sell, while an ask is a price at which one can buy. Mixing asks, stale trades, or nonsynchronous observations does not produce a single frictionless price system. The mathematical inequalities below apply to consistently executable prices under the standing assumptions.

1.5 European and American option price bounds

Source coverage: lecture slides 54–68..

1.5.1 European bounds from prepaid delivery

Set

$$A = A(t,T) = S_t e^{-\delta(T-t)}, \qquad D = D(t,T) = e^{-r(T-t)}.$$

The asset costing A delivers one share at T and the asset costing KD delivers K dollars at T.

Theorem 1.5 (European vanilla bounds)**.** *For a nonnegative underlying with constant dividend yield,*

$$\begin{aligned} \max\{A - KD, 0\} \le c_t \le A, \\ \max\{KD - A, 0\} \le p_t \le KD. \end{aligned}$$

These inequalities do not require $r \ge 0$.

Proof. For every $s \ge 0$ and $K \ge 0$,

$$\begin{aligned} 0 \le (s-K)^+ \le s, \\ s - K \le (s-K)^+, \\ 0 \le (K-s)^+ \le K, \\ K - s \le (K-s)^+. \end{aligned}$$

The claims S_T, K, $S_T - K$, and $K - S_T$ have prices A, KD, $A - KD$, and $KD - A$. Apply payoff dominance to each comparison:

$$\begin{array}{lll} c_t \ge 0, & c_t \ge A - KD, & c_t \le A, \\ p_t \ge 0, & p_t \ge KD - A, & p_t \le KD. \end{array}$$

Combining the lower bounds gives the result. □

The call lower bound as a cash-flow comparison

Compare a call with a long prepaid share and a loan of KD:

Portfolio	Cost at t	$S_T < K$	$S_T \ge K$
Call	c_t	0	$S_T - K$
Prepaid share minus bond	$A - KD$	$S_T - K$	$S_T - K$

The call dominates in every state, so it cannot be cheaper. For a put, reverse the prepaid share and bond: its payoff dominates $K - S_T$.

Remark 1.16 (Sharper upper bounds). The lecture uses $c_t \le S_t$ and, for $r \ge 0$, $p_t \le K$. The prepaid-share and discounted-bond arguments give the sharper European bounds $c_t \le A$ and $p_t \le KD$. When $\delta \ge 0$ and $r \ge 0$, these imply the lecture's bounds.

Example 1.6 (Checking a quote)**.** Let $S_t = 100$, $K = 105$, $T - t = 0.5$, $r = 0.04$, and $\delta = 0.02$. Then

$$\begin{aligned} A &= 100e^{-0.01} \approx 99.0050, \\ KD &= 105e^{-0.02} \approx 102.9209, \\ A - KD &\approx -3.9159. \end{aligned}$$

Thus $0 \le c_t \le 99.0050$ and $3.9159 \le p_t \le 102.9209$. These bounds are necessary; they do not identify a unique option value.

1.5.2 American upper bounds

Theorem 1.6 (American upper and lower bounds)**.** *Assume $r \ge 0$ and $\delta \ge 0$. Then*

$$\begin{aligned} \max\{c_t, (S_t - K)^+\} \le C_t \le S_t, \\ \max\{p_t, (K - S_t)^+\} \le P_t \le K. \end{aligned}$$

In particular,

$$\begin{aligned} S_t \ge C_t \ge c_t \ge \max\{A - KD, 0\}, \\ K \ge P_t \ge p_t \ge \max\{KD - A, 0\}. \end{aligned}$$

Proof. The lower bounds were proved from exercise-right inclusion and immediate exercise.

For the call upper bound, buy one share and sell one American call. If the call is exercised at τ, deliver the share and receive $K \ge 0$. Dividends received before τ are nonnegative and can be banked. If the option expires without exercise, the remaining share and banked dividends have nonnegative value. Therefore this covered position has nonnegative remaining wealth under every exercise decision. If $C_t > S_t$, its setup produces a strictly positive receipt as well, giving arbitrage. Hence $C_t \le S_t$.

For the put upper bound, deposit K in the bank account and sell one American put. At any exercise time τ, the deposit has value $Ke^{r(\tau-t)} \ge K$. Pay K and receive a share worth $S_\tau \ge 0$. Any remaining cash and stock have nonnegative value. If no exercise occurs, keep the bank account. If $P_t > K$, this superhedge also produces an initial receipt, giving arbitrage. Thus $P_t \le K$. □

Remark 1.17 (Where the rate assumption enters). The put proof needs $Ke^{r(\tau-t)} \ge K$. For negative rates, holding K initially may fail to cover a later K payment. A safe general bound for constant r is

$$P_t \le K \max\{1, e^{-r(T-t)}\}.$$

Indeed this initial deposit is at least $Ke^{-r(\tau-t)}$ for every $\tau \in [t, T]$. Similarly, the call hedge assumes the stock position does not require funding negative distributions; nonnegative dividends are sufficient.

1.5.3 Why a non-dividend-paying American call has no early-exercise premium

Theorem 1.7 (American call equals European call)**.** *If the stock pays no dividends and $r \geq 0$, then*

$$C(K,t,T) = c(K,t,T).$$

Early exercise never gives a strictly better value than retaining the option's remaining exercise rights. If $r > 0$, $K > 0$, and $u < T$, exercise of an in-the-money call is strictly dominated by selling the option at its arbitrage-free price.

Proof. First, at any time $u < T$, the European lower bound implies

$$\begin{aligned} c(K,u,T) &\geq S_u - Ke^{-r(T-u)} \\ &= S_u - K + K\left(1 - e^{-r(T-u)}\right) \\ &\geq S_u - K. \end{aligned}$$

It also implies $c(K,u,T) \geq 0$. Consequently,

$$c(K,u,T) \geq (S_u - K)^+.$$

We already know $C_t \geq c_t$. To establish the reverse inequality rigorously, suppose $C_t > c_t$. Buy the European call, sell the American call, and invest the initial receipt $C_t - c_t$. If the short American call is exercised at $\tau < T$, sell the European call. Its market value satisfies

$$c(K,\tau,T) \geq (S_\tau - K)^+,$$

so the sale funds the short contract's exercise obligation. For physical settlement, the stock can be bought at S_τ and delivered against receipt of K; the net cost is $S_\tau - K$, also covered by the same bound. Bank any surplus. If exercise is deferred to T, the two terminal payoffs cancel. In either case the invested initial receipt remains strictly positive. This would be arbitrage, so $C_t \leq c_t$.

For $r > 0$, $K > 0$, and $u < T$, the term $K(1 - e^{-r(T-u)})$ is strictly positive. If $S_u > K$, the option's price is therefore strictly above the exercise receipt $S_u - K$. □

Remark 1.18 (Weak dominance and investor preferences)*.* At $r = 0$ or $K = 0$, indifference can occur, so "never exercise early" should be read as "early exercise is not strictly necessary to obtain the option's value". An investor may still wish to sell the option before expiry. This theorem concerns the value of the exercise right, not an instruction to maintain market exposure until T.

Remark 1.19 (Why dividends change the conclusion)*.* For $\delta > 0$, the lower bound becomes $S_u e^{-\delta(T-u)} - Ke^{-r(T-u)}$, which need not exceed $S_u - K$. Exercising acquires the stock and its future dividend rights. Therefore early exercise can have value; positive dividends do not imply that early exercise is always optimal.

Example 1.7 (Negative rates invalidate the no-early-exercise conclusion)**.** Consider a deterministic non-dividend stock $S_u = S_t e^{r(u-t)}$ with $r < 0$, and suppose $S_T > K$. The European payoff is deterministic, so

$$\begin{aligned} c_t &= e^{-r(T-t)}(S_t e^{r(T-t)} - K) \\ &= S_t - Ke^{-r(T-t)} \\ &< S_t - K. \end{aligned}$$

Immediate American exercise is worth $S_t - K$, exceeding the European value. This market has no stock-versus-bank arbitrage: the stock grows at the same rate as the bank account.

1.5.4 American puts and early exercise

Example 1.8 (Early exercise can be optimal for a put)**.** Suppose the stock is worthless now and remains zero, and $r > 0$, $K > 0$. Exercising an American put now gives K. Waiting until T gives a payment with present value $Ke^{-r(T-t)} < K$. Thus

$$\begin{aligned} P_t &= K, \\ p_t &= Ke^{-r(T-t)}, \\ P_t - p_t &= K(1 - e^{-r(T-t)}) > 0. \end{aligned}$$

This proves that an American put can have a strictly positive early-exercise premium. It does not prove that every American put should be exercised early.

Proposition 1.7 (No put exercise premium at zero rates under nonnegative yield)**.** *If $r = 0$ and $\delta \geq 0$, then $P_t = p_t$.*

Proof. At every possible exercise time u, the European put lower bound gives

$$\begin{aligned} p(K, u, T) &\geq \max\{K - S_u e^{-\delta(T-u)}, 0\} \\ &\geq (K - S_u)^+. \end{aligned}$$

A long European put can therefore be sold to cover any exercise obligation on a short American put. If $P_t > p_t$, buy the European, sell the American, and invest the difference to obtain arbitrage, exactly as in the call proof. Since $P_t \geq p_t$, equality follows. □

Remark 1.20. The lecture's statement about possible early put exercise should not be interpreted as a universal optimal-exercise rule. Positive rates encourage earlier receipt of the strike, while retaining optionality and receiving the benefit of future price declines can encourage waiting. Part I supplies comparisons, not a general exercise boundary.

1.6 Put-call parity and its American bounds

Source coverage: lecture slides 69–76..

1.6.1 European parity

Theorem 1.8 (Put-call parity)**.** *For European options on the same underlying with the same strike K and maturity T,*

$$\boxed{c_t - p_t = A(t,T) - KD(t,T).}$$

With continuous proportional dividends,

$$c_t - p_t = S_t e^{-\delta(T-t)} - Ke^{-r(T-t)}.$$

Proof. At maturity,

$$\begin{aligned}(S_T - K)^+ - (K - S_T)^+ &= \begin{cases}(S_T - K) - 0, & S_T \geq K,\\ 0 - (K - S_T), & S_T < K\end{cases}\\ &= S_T - K.\end{aligned}$$

Thus long call minus long put replicates long prepaid share minus a bond paying K. The two costs must coincide:

$$c_t - p_t = A(t,T) - KD(t,T).$$

□

Equivalently,

$$\begin{aligned}c_t + KD &= p_t + A,\\ c_t &= p_t + A - KD,\\ p_t &= c_t - A + KD.\end{aligned}$$

The first identity says that a call plus enough cash to pay the strike is equivalent to a prepaid share protected by a put.

Corollary 1.2 (Synthetic forward)**.** *A long call and short put with common K, T have exactly the value and payoff of the long forward with delivery price K:*

$$c_t - p_t = f_t(K,T).$$

This synthetic forward has zero cost only if $K = F(t,T)$.

Corollary 1.3 (Discrete-dividend parity)**.** *For known discrete dividends with present value I_t,*

$$c_t - p_t = S_t - I_t - KD(t,T).$$

Proof. The payoff identity is unchanged; replace the prepaid-share price by $S_t - I_t$. □

1.6.2 Explicit arbitrage for a parity violation

Let

$$\varepsilon = (c_t - p_t) - (A - KD).$$

If $\varepsilon > 0$, sell the call, buy the put, buy prepaid stock, and borrow KD. The setup cash flow is

$$c_t - p_t - A + KD = \varepsilon > 0,$$

and the terminal cash flow before banking this receipt is

$$\begin{aligned} -(S_T - K)^+ + (K - S_T)^+ + S_T - K &= -(S_T - K) + S_T - K \\ &= 0. \end{aligned}$$

Investing the initial receipt yields terminal profit $\varepsilon e^{r(T-t)}$. If $\varepsilon < 0$, reverse every position and bank $-\varepsilon$.

Example 1.9 (The lecture's numerical parity trade)**.** Treat the quoted instruments as European and assume no dividends. Use the lecture's stylized data

$$S_0 = 245, \quad K = 250, \quad T = \frac{1}{52}, \quad r = 0.02, \quad c_0 = 3.20, \quad p_0 = 8.10.$$

Then

$$\begin{aligned} c_0 - p_0 &= -4.90, \\ S_0 - Ke^{-rT} &= 245 - 250e^{-0.02/52} \\ &\approx -4.903864643, \\ \varepsilon &= -4.90 - (-4.903864643) \\ &\approx 0.003864643 > 0. \end{aligned}$$

Sell one call, buy one put, buy one share, and borrow $250e^{-0.02/52}$. The complete cash-flow table is

Position	At 0	At T, $S_T < 250$	At T, $S_T \geq 250$
Buy stock	-245	S_T	S_T
Borrow	$+249.903864643$	-250	-250
Sell call	$+3.20$	0	$-(S_T - 250)$
Buy put	-8.10	$250 - S_T$	0
Total	$+0.003864643$	0	0

Banking the receipt makes the strategy zero-cost and produces approximately 0.003866129 at T per share of exposure.

Remark 1.21 (The example's modelling status). The slide data are historical and rounded. The exact-parity trade assumes European exercise, synchronous executable prices, no dividends, and accessible 2% financing. American equity-option quotes cannot simply be substituted into European parity without checking exercise features. A tiny algebraic discrepancy in indicative quotes is not evidence of an executable real-market arbitrage.

1.6.3 American put-call inequalities

Theorem 1.9 (American parity bounds without dividends)**.** *Assume no dividends and $r \geq 0$. Then*

$$\boxed{S_t - K \leq C_t - P_t \leq S_t - KD(t,T).}$$

Proof. For the upper bound, $C_t = c_t$ and $P_t \geq p_t$. Therefore

$$\begin{aligned} C_t - P_t &\leq c_t - p_t \\ &= S_t - KD(t,T). \end{aligned}$$

For the lower bound, construct a portfolio consisting of one long American call, K dollars in the bank, one short American put, and one short share. Its setup cost is

$$C_t + K - P_t - S_t.$$

If the short put is exercised at time τ, pay K from the bank and receive the share. Use that share to close the short stock position. The cash remaining is

$$Ke^{r(\tau - t)} - K \geq 0,$$

and the unexercised long call has nonnegative value. If the put is not exercised early, at maturity the call, put, and stock positions combine as

$$\begin{aligned} (S_T - K)^+ - (K - S_T)^+ - S_T &= (S_T - K) - S_T \\ &= -K. \end{aligned}$$

The bank account covers this with $Ke^{r(T-t)} \geq K$. Thus the portfolio never requires outside funding and has nonnegative residual value. A negative initial cost would yield arbitrage after banking the receipt. Hence

$$\begin{aligned} C_t + K - P_t - S_t &\geq 0, \\ C_t - P_t &\geq S_t - K. \end{aligned}$$

□

Corollary 1.4 (Bound on the put's early-exercise premium)**.** *Under the same assumptions,*

$$0 \leq P_t - p_t \leq K(1 - D(t,T)).$$

Proof. Use $C_t = c_t$, $c_t - p_t = S_t - KD$, and $C_t - P_t \geq S_t - K$:

$$\begin{aligned} S_t - KD - (P_t - p_t) &\geq S_t - K, \\ P_t - p_t &\leq K - KD. \end{aligned}$$

The lower bound follows from the American exercise advantage. □

Remark 1.22 (The gap is not universally small). The width of the American parity interval is $K(1 - e^{-r(T-t)})$. It is approximately $Kr(T-t)$ for a small product $r(T-t)$, but it can be economically large for long maturities, high rates, or large strikes.

1.7 Dependence on strike: monotonicity, Lipschitz bounds, and convexity

Source coverage: lecture slides 77–86.. Throughout this section, fix t, T, the underlying, and the exercise style. Abbreviate $c(K) = c(K, t, T)$, and similarly for p, C, P. All comparisons require the same underlying and maturity.

1.7.1 European strike spreads

Lemma 1.3 (Pointwise strike-difference bounds)**.** *For $0 \le K_1 \le K_2$, let $d = K_2 - K_1$. For every $s \ge 0$,*

$$0 \le (s - K_1)^+ - (s - K_2)^+ \le d,$$
$$0 \le (K_2 - s)^+ - (K_1 - s)^+ \le d.$$

Proof. The call difference is

$$(s - K_1)^+ - (s - K_2)^+ = \begin{cases} 0, & s \le K_1, \\ s - K_1, & K_1 < s < K_2, \\ K_2 - K_1, & s \ge K_2. \end{cases}$$

Every branch lies between zero and d. The put difference is

$$(K_2 - s)^+ - (K_1 - s)^+ = \begin{cases} K_2 - K_1, & s \le K_1, \\ K_2 - s, & K_1 < s < K_2, \\ 0, & s \ge K_2, \end{cases}$$

and the same bounds follow. □

Theorem 1.10 (European monotonicity and discounted Lipschitz bounds)**.** *For $K_1 \le K_2$,*

$$0 \le c(K_1) - c(K_2) \le D(t, T)(K_2 - K_1),$$
$$0 \le p(K_2) - p(K_1) \le D(t, T)(K_2 - K_1).$$

Thus European calls are nonincreasing in strike, and European puts are nondecreasing in strike. Both are $D(t, T)$-Lipschitz.

Proof. Each difference portfolio has terminal payoff between zero and d. The price comparison principle places its price between zero and $dD(t, T)$. □

Remark 1.23 (Recovering the lecture's 1-Lipschitz statement). If $r \ge 0$, then $D(t, T) \le 1$. Consequently,

$$|c(K_2) - c(K_1)| \le |K_2 - K_1|,$$
$$|p(K_2) - p(K_1)| \le |K_2 - K_1|.$$

For negative r, the discounted constant can exceed one, so the 1-Lipschitz claim is not generally valid. The discounted bounds remain valid.

1.7.2 American strike comparisons: every exercise time must be covered

Theorem 1.11 (American monotonicity and 1-Lipschitz bounds)**.** *Assume $r \geq 0$. For $K_1 \leq K_2$,*

$$0 \leq C(K_1) - C(K_2) \leq K_2 - K_1,$$
$$0 \leq P(K_2) - P(K_1) \leq K_2 - K_1.$$

Proof. Write $d = K_2 - K_1$. We prove each inequality with a covering portfolio.

Call monotonicity. Buy a call at K_1 and sell a call at K_2. If the short option is exercised at τ, the pointwise inequality

$$(S_\tau - K_1)^+ \geq (S_\tau - K_2)^+$$

shows that the long option can cover its obligation by exercise. If the short expires unexercised, the long position still has nonnegative payoff. Therefore $C(K_1) - C(K_2) \geq 0$.

Call spread upper bound. Buy the call at K_2, sell the call at K_1, and place d dollars in the bank. At the exercise time τ of the short option, the bank has $de^{r(\tau-t)} \geq d$. The pointwise spread bound implies

$$\begin{aligned}(S_\tau - K_2)^+ + de^{r(\tau-t)} &\geq (S_\tau - K_2)^+ + d \\ &\geq (S_\tau - K_1)^+.\end{aligned}$$

Exercise the long option if useful and use the cash to settle the short. If the short is never exercised, the remaining assets are nonnegative. Therefore the superhedge cannot have negative cost:

$$C(K_2) + d - C(K_1) \geq 0,$$
$$C(K_1) - C(K_2) \leq d.$$

Put monotonicity. Buy a put at K_2 and sell a put at K_1. At any exercise time of the short,

$$(K_2 - S_\tau)^+ \geq (K_1 - S_\tau)^+.$$

Thus the long covers the short and $P(K_2) - P(K_1) \geq 0$.

Put spread upper bound. Buy a put at K_1, sell a put at K_2, and bank d. At exercise,

$$\begin{aligned}(K_1 - S_\tau)^+ + de^{r(\tau-t)} &\geq (K_1 - S_\tau)^+ + d \\ &\geq (K_2 - S_\tau)^+.\end{aligned}$$

The same covering argument gives $P(K_1) + d - P(K_2) \geq 0$, or $P(K_2) - P(K_1) \leq d$. □

Remark 1.24 (Cash settlement is a convenient representation)*.* A physical call exercise can be settled by buying a share at S_τ, delivering it, and receiving K; the net cost is $S_\tau - K$. A physical put exercise can be settled by paying K, receiving the share, and selling it for S_τ; the net cost is $K - S_\tau$. These transactions justify using exercise cash values in the covering arguments. An irrational out-of-the-money exercise by the short holder benefits the writer and cannot invalidate the bounds.

Remark 1.25 (A negative-rate extension)*.* For constant $r < 0$, deposit $de^{-r(T-t)}$ rather than d. This covers a payment of at most d at any exercise date. Thus an always-valid bound in this financing model is

$$|C(K_2) - C(K_1)|,\ |P(K_2) - P(K_1)| \leq d \max\{1, D(t,T)\}.$$

Monotonicity itself does not require nonnegative rates.

1.7.3 Convexity in the strike

Definition 1.22 (Convex function)**.** A function g on an interval is convex if, for every x, y in that interval and $\lambda \in [0,1]$,

$$g(\lambda x + (1-\lambda)y) \le \lambda g(x) + (1-\lambda)g(y).$$

Geometrically, the graph lies below each chord joining two of its points.

Theorem 1.12 (Convexity of vanilla prices in strike)**.** *Each of c, p, C, P is convex as a function of strike. In particular, writing $\bar{K} = \lambda K_1 + (1-\lambda)K_2$,*

$$\begin{aligned} c(\bar{K}) &\le \lambda c(K_1) + (1-\lambda)c(K_2), \\ p(\bar{K}) &\le \lambda p(K_1) + (1-\lambda)p(K_2), \\ C(\bar{K}) &\le \lambda C(K_1) + (1-\lambda)C(K_2), \\ P(\bar{K}) &\le \lambda P(K_1) + (1-\lambda)P(K_2). \end{aligned}$$

Proof. For a fixed stock value s,

$$\begin{aligned} (s-\bar{K})^+ &= \big(\lambda(s-K_1) + (1-\lambda)(s-K_2)\big)^+ \\ &\le \lambda(s-K_1)^+ + (1-\lambda)(s-K_2)^+, \\ (\bar{K}-s)^+ &= \big(\lambda(K_1-s) + (1-\lambda)(K_2-s)\big)^+ \\ &\le \lambda(K_1-s)^+ + (1-\lambda)(K_2-s)^+. \end{aligned}$$

For European options, buy the indicated weighted endpoint options and sell one option at $\bar{K}$. The terminal payoff is nonnegative, so the initial cost is nonnegative. This proves the first two inequalities.

For American options, buy the same weighted endpoint options and sell the middle-strike option. At the exercise time τ of the short option, exercise those long positions with positive exercise value. The pointwise inequalities, with $s = S_\tau$, show that their combined receipts cover the obligation. Any unused long rights can be retained or discarded at nonnegative value. If the short is not exercised before expiry, the same comparison works at T. Thus the portfolio is a superhedge under every exercise time, so its cost is nonnegative. This proves the remaining two inequalities. □

Explicit piecewise call-butterfly calculation

Assume $K_1 < K_2$ and $0 < \lambda < 1$, so $K_1 < \bar{K} < K_2$. The weighted payoff

$$b(s) = \lambda(s-K_1)^+ + (1-\lambda)(s-K_2)^+ - (s-\bar{K})^+$$

is

$$b(s) = \begin{cases} 0, & s \le K_1, \\ \lambda(s-K_1), & K_1 < s < \bar{K}, \\ (1-\lambda)(K_2-s), & \bar{K} \le s < K_2, \\ 0, & s \ge K_2. \end{cases}$$

For the third branch, the algebra is

$$\begin{aligned} \lambda(s-K_1) - (s-\bar{K}) &= \lambda s - \lambda K_1 - s + \lambda K_1 + (1-\lambda)K_2 \\ &= (1-\lambda)(K_2-s) \ge 0. \end{aligned}$$

For the fourth branch,

$$\begin{aligned}\lambda(s-K_1)+(1-\lambda)(s-K_2)-(s-\bar{K}) &= s-\lambda K_1-(1-\lambda)K_2-s+\bar{K}\\ &= 0.\end{aligned}$$

The put version has exactly the same weighted payoff: use $(K-s)^+=(s-K)^++K-s$, and observe that the affine strike terms cancel.

1.7.4 The lecture's convexity arbitrage

Example 1.10 (Unequally spaced strikes)**.** Suppose European call quotes with common maturity are

$$c(50)=14, \qquad c(59)=8.9, \qquad c(65)=5.$$

Solve for the endpoint weight:

$$\begin{aligned}59 &= \lambda 50+(1-\lambda)65,\\ 15\lambda &= 6,\\ \lambda &= 0.4.\end{aligned}$$

Convexity requires

$$\begin{aligned}c(59) &\le 0.4c(50)+0.6c(65)\\ &= 0.4(14)+0.6(5)=8.6.\end{aligned}$$

The quoted 8.9 violates this bound. Buy four calls at 50, buy six calls at 65, and sell ten calls at 59. The initial cash flow is

$$\begin{aligned}-4(14)-6(5)+10(8.9) &= -56-30+89\\ &= 3.\end{aligned}$$

The terminal payoff is

$$\begin{aligned}H(s) &= 4(s-50)^+ + 6(s-65)^+ - 10(s-59)^+\\ &= \begin{cases}0, & s\le 50,\\ 4(s-50), & 50<s\le 59,\\ 4(s-50)-10(s-59), & 59<s<65,\\ 4(s-50)+6(s-65)-10(s-59), & s\ge 65\end{cases}\\ &= \begin{cases}0, & s\le 50,\\ 4(s-50), & 50<s\le 59,\\ 6(65-s), & 59<s<65,\\ 0, & s\ge 65.\end{cases}\end{aligned}$$

This is nonnegative everywhere and reaches 36 at $s=59$. Bank the initial three-dollar receipt. The zero-cost strategy then has wealth $H(S_T)+3e^{r(T-t)}>0$ in every state.

1.7.5 Slopes, continuity, and discrete quote checks

Corollary 1.5 (Continuity and slope restrictions)**.** *European call and put prices are continuous in strike. Wherever the derivative exists,*

$$-D(t,T) \le c'(K) \le 0,$$
$$0 \le p'(K) \le D(t,T).$$

For American options with $r \ge 0$, the corresponding bounds are $-1 \le C'(K) \le 0$ and $0 \le P'(K) \le 1$. Convexity implies that secant slopes increase from left to right.

Proof. The Lipschitz bounds imply continuity directly: if $K_n \to K$, then $|c(K_n) - c(K)| \le D|K_n - K| \to 0$, and likewise for the other functions. Divide each strike-difference inequality by $K_2 - K_1 > 0$ and take a derivative limit where it exists.

For the secant statement, take $K_1 < K_2 < K_3$ and write

$$K_2 = \frac{K_3 - K_2}{K_3 - K_1} K_1 + \frac{K_2 - K_1}{K_3 - K_1} K_3.$$

For any convex price function g, convexity gives

$$(K_3 - K_1)g(K_2) \le (K_3 - K_2)g(K_1) + (K_2 - K_1)g(K_3),$$
$$(K_3 - K_2)[g(K_2) - g(K_1)] \le (K_2 - K_1)[g(K_3) - g(K_2)].$$

Divide by $(K_2 - K_1)(K_3 - K_2) > 0$ to obtain

$$\frac{g(K_2) - g(K_1)}{K_2 - K_1} \le \frac{g(K_3) - g(K_2)}{K_3 - K_2}.$$

□

Remark 1.26 (A practical mathematical check). For a finite European call grid, adjacent secant slopes must lie in $[-D, 0]$ and be nondecreasing. For puts they must lie in $[0, D]$ and be nondecreasing. These are useful necessary checks, alongside individual bounds and parity. They are not, by themselves, a complete test of consistency across all maturities and all traded instruments.

1.8 Dependence on maturity

Source coverage: lecture slides 87–88..

1.8.1 American options with more time

Theorem 1.13 (American maturity monotonicity)**.** *For $t \le T_1 \le T_2$, with otherwise identical contract terms,*

$$C(K,t,T_2) \ge C(K,t,T_1),$$
$$P(K,t,T_2) \ge P(K,t,T_1).$$

Proof. Buy the longer-maturity option and sell the shorter-maturity option. Whenever the short option is exercised at $\tau \le T_1$, exercise the long option simultaneously. Its exercise date is permitted because $\tau \le T_1 \le T_2$, and the exercise receipts cancel the short obligation. If the short expires unexercised, the remaining long option has nonnegative value. Thus the position cannot have negative initial cost without arbitrage. This proves each inequality. □

Remark 1.27. The proof uses inclusion of exercise opportunities, not a comparison of the stock price at two different dates. It does not require $r \ge 0$.

1.8.2 Why European options need additional conditions

European options with different maturities pay at different dates and depend on different stock values. A later European expiry does not include the earlier exercise right.

Example 1.11 (A longer European put can be cheaper)**.** Suppose $S_u = 0$ for all $u \ge t$, with $r > 0$ and $K > 0$. Then

$$\begin{aligned} p(K,t,T_1) &= Ke^{-r(T_1-t)}, \\ p(K,t,T_2) &= Ke^{-r(T_2-t)} \\ &= p(K,t,T_1)e^{-r(T_2-T_1)} \\ &< p(K,t,T_1) \qquad (T_2 > T_1). \end{aligned}$$

Both puts pay the same amount K, but the later payment is worth less now.

Example 1.12 (A longer European call can be cheaper with dividends)**.** Let $r = 0$, $\delta > 0$, and suppose the ex-dividend stock evolves deterministically as $S_u = S_t e^{-\delta(u-t)}$. Reinvested total stock wealth is constant, consistent with the zero-rate bank account. Choose $T_1 < T_2$ with $S_{T_2} > K$. Then the deterministic European call values are

$$\begin{aligned} c(K,t,T_1) &= S_t e^{-\delta(T_1-t)} - K, \\ c(K,t,T_2) &= S_t e^{-\delta(T_2-t)} - K \\ &< c(K,t,T_1). \end{aligned}$$

Thus there is no unconditional maturity monotonicity for European calls when dividends are allowed.

Corollary 1.6 (An important European-call exception)**.** *For a non-dividend-paying stock and* $r \geq 0$*,*

$$c(K, t, T_2) \geq c(K, t, T_1) \qquad (T_2 \geq T_1).$$

Proof. For each maturity, American and European call values coincide. Hence

$$\begin{aligned} c(K, t, T_2) &= C(K, t, T_2) \\ &\geq C(K, t, T_1) \\ &= c(K, t, T_1). \end{aligned}$$

□

Remark 1.28. The lecture's warning that European maturity comparisons need not hold is correct in general. It should not obscure this useful special case, which follows from results already proved in Part I.

1.9 Engineering payoffs with stocks, bonds, and options

Source coverage: lecture slides 89–106.. For this section, all options are European with a common expiry T. Unless stated otherwise, assume no dividends, so a long share contributes terminal wealth S_T without additional intermediate cash flows. With dividends, the formulas still describe the terminal ex-dividend positions, but total profit must also account for all distributions and financing. A prepaid share can be used when a terminal-only share claim is intended.

Write $s = S_T$. Every displayed payoff is a terminal value *before* subtracting financed setup cost. Negative initial cost means an initial credit, which must still be compared with future liabilities.

1.9.1 Protective puts: a floor on terminal value

Definition 1.23 (Protective put or floor)**.** A protective put consists of one long share and one long put with strike K.

Proposition 1.8 (Floor payoff and replication)**.** *The protective put has payoff*

$$s + (K - s)^+ = \max\{s, K\} = K + (s - K)^+.$$

Its price is $S_t + p_t = KD + c_t$ in the no-dividend setting.

Proof. If $s < K$, the put adds $K - s$, so the total is K. If $s \geq K$, the put pays zero, so the total is s. Also,

$$s + (K - s)^+ = K + (s - K)^+$$

by the positive-part identity. The right side is replicated by a K-face-value bond and a call. Equality of prices follows from no-arbitrage or put-call parity. □

If setup cost is $v = S_t + p_t$, financed profit satisfies

$$\Pi_T \geq K - ve^{r(T-t)}.$$

The floor is on terminal portfolio value, not a guarantee of nonnegative profit. Insurance has a cost.

1.9.2 Covered calls: selling the upper tail

Definition 1.24 (Covered call)**.** A covered call consists of one long share and one short call on that share.

Proposition 1.9 (Covered-call payoff)**.**

$$\begin{aligned} s - (s - K)^+ &= \min\{s, K\} \\ &= K - (K - s)^+. \end{aligned}$$

Consequently its cost satisfies $S_t - c_t = KD - p_t$.

Proof. For $s \leq K$, the call pays zero and the position is worth s. For $s > K$, the call liability is $s - K$, leaving K. The alternative expression follows from the positive-part identity, and prices follow by replication. □

Remark 1.29 (Premium is not guaranteed profit). Selling the call produces a known initial receipt, but the stock can still fall to zero. The payoff is capped above at K and remains exposed to stock losses below K. Describing the premium as income does not remove the portfolio's risk or financing costs.

1.9.3 Protecting and covering short stock

Definition 1.25 (Cap on a short position)**.** In the lecture's terminology, a cap consists of short stock plus a long call. Its terminal payoff is

$$-s + (s - K)^+ = -\min\{s, K\}.$$

For $s < K$, the payoff is $-s$; for $s \geq K$, it is $-K$. Thus the terminal cost of closing the short exposure is capped at K. Initial short-sale proceeds and the call premium must be included to obtain total profit.

Definition 1.26 (Covered put)**.** A covered put consists of short stock plus a short put. Its terminal payoff is

$$-s - (K - s)^+ = -\max\{s, K\}.$$

For $s < K$, the payoff is $-K$; for $s \geq K$, it is $-s$. The premium compensates for giving up favourable outcomes from further stock-price declines, but unbounded loss from a large rise in the stock remains.

Remark 1.30 (Terminology). Here "floor" and "cap" refer to the stock-option combinations in the lecture. In other settings the same terms describe interest-rate derivatives; those are different contracts.

1.9.4 Bull and bear spreads with calls and puts

Fix $K_1 < K_2$ and $d = K_2 - K_1$.

Definition 1.27 (Bull call spread)**.** Buy one call at K_1 and sell one call at K_2. The payoff is

$$H_{\mathrm{BC}}(s) = (s - K_1)^+ - (s - K_2)^+ = \begin{cases} 0, & s \leq K_1, \\ s - K_1, & K_1 < s < K_2, \\ d, & s \geq K_2. \end{cases}$$

This payoff increases with s but is capped. Its cost is $v_{\mathrm{BC}} = c(K_1) - c(K_2)$, with $0 \leq v_{\mathrm{BC}} \leq dD$. Let $a = v_{\mathrm{BC}} e^{r(T-t)} \in [0, d]$. Then

$$\begin{aligned} \min_s \Pi_T &= -a, \\ \max_s \Pi_T &= d - a, \\ \text{interior break-even} &= K_1 + a. \end{aligned}$$

If $a = 0$ or $a = d$, a whole flat region may also have zero profit.

Definition 1.28 (Bear put spread)**.** Buy one put at K_2 and sell one put at K_1. The payoff is

$$H_{\mathrm{BP}}(s) = (K_2 - s)^+ - (K_1 - s)^+ = \begin{cases} d, & s \le K_1, \\ K_2 - s, & K_1 < s < K_2, \\ 0, & s \ge K_2. \end{cases}$$

The cost is $v_{\mathrm{BP}} = p(K_2) - p(K_1)$, again between zero and dD. With $a = v_{\mathrm{BP}} e^{r(T-t)}$, the minimum and maximum financed profits are $-a$ and $d - a$, and the interior break-even is $K_2 - a$.

The other two directional spreads

A bear call spread sells the lower-strike call and buys the higher-strike call; its payoff is $-H_{\mathrm{BC}}$. A bull put spread buys the lower-strike put and sells the higher-strike put; its payoff is $-H_{\mathrm{BP}}$.

Proposition 1.10 (Call and put spreads differ by a bond)**.** *For every s,*

$$\begin{aligned} H_{\mathrm{BC}}(s) + H_{\mathrm{BP}}(s) &= d, \\ H_{\mathrm{BC}}(s) - \left[(K_1 - s)^+ - (K_2 - s)^+\right] &= d. \end{aligned}$$

Consequently,

$$c(K_1) - c(K_2) + p(K_2) - p(K_1) = dD.$$

Proof. Subtract the positive-part identities at K_1 and K_2:

$$\begin{aligned} &(s - K_1)^+ - (K_1 - s)^+ \\ &\quad - \left[(s - K_2)^+ - (K_2 - s)^+\right] \\ &= (s - K_1) - (s - K_2) = K_2 - K_1 = d. \end{aligned}$$

Rearrange and apply the law of one price. Thus a bull call spread equals a bull put spread plus a bond paying d. The two have different initial costs but, after consistent financing, the same profit profile. □

1.9.5 Collars: imposing both a floor and a cap

Definition 1.29 (Collar)**.** A collar consists of one long share, one long put at K_L, and one short call at K_U, where $K_L < K_U$.

Proposition 1.11 (Collar payoff and decomposition)**.**

$$\begin{aligned} H_{\mathrm{col}}(s) &= s + (K_L - s)^+ - (s - K_U)^+ \\ &= \begin{cases} K_L, & s \le K_L, \\ s, & K_L < s < K_U, \\ K_U, & s \ge K_U \end{cases} \\ &= K_L + (s - K_L)^+ - (s - K_U)^+. \end{aligned}$$

Its cost is

$$S_t + p(K_L) - c(K_U) = K_L D + c(K_L) - c(K_U).$$

Proof. If $s \le K_L$, the put restores stock value to K_L, while the call is worthless. If $K_L < s < K_U$, both options are worthless. If $s \ge K_U$, the call removes all stock value above K_U. The last expression follows from $s + (K_L - s)^+ = K_L + (s - K_L)^+$. Its price is the cost of a bond paying K_L plus a bull call spread. □

Remark 1.31 (Zero-cost collars). If $p(K_L) = c(K_U)$, the *option overlay* has zero initial cost. The stock still costs S_t. A zero-cost overlay is therefore not a zero-cost total investment.

Remark 1.32 (What no-arbitrage does and does not say). A collar can reduce downside exposure and change return variability. No-arbitrage alone does not prohibit an improvement in a chosen risk-adjusted performance statistic, nor does it specify expected returns. What it prohibits is a freely financed strategy with no losses and a genuine gain. The lecture's historical anecdote is not a mathematical proof about risk-adjusted performance.

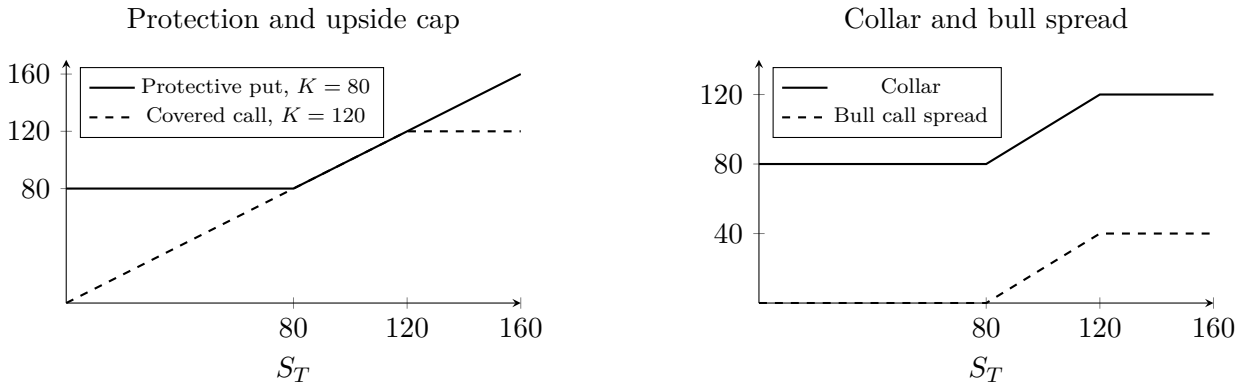


Figure 1.2: Exact terminal payoff profiles. The collar with strikes 80 and 120 equals the bull spread plus 80 dollars at maturity.

1.9.6 Straddles and strangles

Definition 1.30 (Long straddle)**.** A long straddle buys a call and a put with the same strike K and expiry. An at-the-money straddle chooses K near the current spot price, but the payoff identity does not require that choice.

Proposition 1.12 (Straddle payoff and break-even)**.** *The terminal payoff is*

$$\begin{aligned} H_{\text{strad}}(s) &= (s - K)^+ + (K - s)^+ \\ &= |s - K|. \end{aligned}$$

If its premium is $v = c(K) + p(K)$ and $a = ve^{r(T-t)}$, its financed profit is $|s - K| - a$. Break-even values are $K - a$ and $K + a$ whenever they lie in the admissible domain $s \ge 0$.

Proof. Use $x^+ + (-x)^+ = |x|$ with $x = s - K$. Solve $|s - K| = a$ to obtain the two break-even values. If $a > K$, the lower value is negative and cannot be attained by a nonnegative stock. □

The largest loss occurs at $s = K$ and equals the financed premium a. Profit requires a sufficiently large terminal deviation, not merely an intraday fluctuation that reverses before expiry.

Definition 1.31 (Long strangle)**.** A long strangle buys a put at K_L and a call at K_U, where $K_L < K_U$. A usual out-of-the-money strangle has $K_L < S_t < K_U$.

Its payoff is

$$\begin{aligned} H_{\text{strang}}(s) &= (K_L - s)^+ + (s - K_U)^+ \\ &= \begin{cases} K_L - s, & s < K_L, \\ 0, & K_L \le s \le K_U, \\ s - K_U, & s > K_U. \end{cases} \end{aligned}$$

Let $v = p(K_L) + c(K_U)$ and $a = ve^{r(T-t)}$. Break-even occurs at $K_L - a$ if nonnegative, and at $K_U + a$. The entire interval $[K_L, K_U]$ produces the maximum loss a.

Remark 1.33 (The meaning of "long volatility"). The straddle and strangle payoffs are convex functions of s. They benefit from sufficiently large terminal moves relative to their premiums. In a pricing model that relates greater dispersion to convex-payoff values, this motivates the phrase "long volatility". Part I has not yet introduced such a model: a payoff diagram alone does not establish a sensitivity to every volatility parameter or a response to every change in realized path variation.

1.9.7 Butterflies and the lecture's shifted convention

Definition 1.32 (Standard symmetric long call butterfly)**.** Let $K_L < K_M < K_U$ with $K_M - K_L = K_U - K_M = h > 0$. Buy one call at K_L, sell two calls at K_M, and buy one call at K_U.

Proposition 1.13 (Butterfly payoff)**.** *The standard butterfly payoff is*

$$\begin{aligned} B(s) &= (s - K_L)^+ - 2(s - K_M)^+ + (s - K_U)^+ \\ &= \begin{cases} 0, & s \le K_L, \\ s - K_L, & K_L < s \le K_M, \\ K_U - s, & K_M < s < K_U, \\ 0, & s \ge K_U. \end{cases} \end{aligned}$$

In particular, $0 \le B(s) \le h$. *Its cost* $b = c(K_L) - 2c(K_M) + c(K_U)$ *satisfies* $0 \le b \le hD$.

Proof. For $s \le K_L$, every call is worthless. For $K_L < s \le K_M$, only the first call contributes. For $K_M < s < K_U$,

$$\begin{aligned} B(s) &= (s - K_L) - 2(s - K_M) \\ &= 2K_M - K_L - s \\ &= K_U - s. \end{aligned}$$

For $s \ge K_U$,

$$\begin{aligned} B(s) &= s - K_L - 2s + 2K_M + s - K_U \\ &= 2K_M - K_L - K_U = 0. \end{aligned}$$

The peak at K_M is h. Apply payoff bounds to obtain the cost bounds. Convexity independently gives $b \ge 0$. □

With $a = be^{r(T-t)} \in [0, h]$, financed profit is $B(s) - a$. Maximum loss is a, maximum profit is $h - a$, and the two interior break-even points are $K_L + a$ and $K_U - a$.

Proposition 1.14 (Short straddle plus long strangle is a shifted butterfly)**.** *The lecture's combination is*

$$L(s) = (K_L - s)^+ + (s - K_U)^+ - (s - K_M)^+ - (K_M - s)^+.$$

For symmetric strikes,

$$L(s) = B(s) - h.$$

Thus it has payoff $-h$ in both tails and zero at the middle strike. It is not the nonnegative standard butterfly payoff.

Proof. Use $(K - s)^+ = (s - K)^+ + K - s$ twice:

$$\begin{aligned} L(s) &= (s - K_L)^+ + K_L - s + (s - K_U)^+ \\ &\quad - (s - K_M)^+ - \big((s - K_M)^+ + K_M - s\big) \\ &= (s - K_L)^+ - 2(s - K_M)^+ + (s - K_U)^+ + K_L - K_M \\ &= B(s) - h. \end{aligned}$$

Its initial value is consequently $\ell = b - hD$. After consistent financing,

$$\begin{aligned} L(s) - \ell e^{r(T-t)} &= B(s) - h - (b - hD)e^{r(T-t)} \\ &= B(s) - h - be^{r(T-t)} + h \\ &= B(s) - be^{r(T-t)}. \end{aligned}$$

The shifted construction and standard butterfly therefore have identical financed profit, despite different payoff levels and initial prices. □

Example 1.13 (Strikes 80, 100, and 120)**.** Here $h = 20$. The standard butterfly pays zero outside $[80, 120]$ and 20 at $s = 100$. The lecture's short-straddle-plus-long-strangle combination pays -20 outside that interval and zero at $s = 100$. Adding a bond paying 20 converts the lecture's payoff into the standard one.

Remark 1.34 (Butterflies are not globally concave). The tent-shaped payoff rises and then falls, but its slope also jumps upward at the outer strikes. It is therefore not globally concave. A butterfly concentrates its favourable outcomes near the middle strike; describing it as "short volatility" is a qualitative description near that region, not a universal sign theorem for volatility sensitivity across all prices and models.

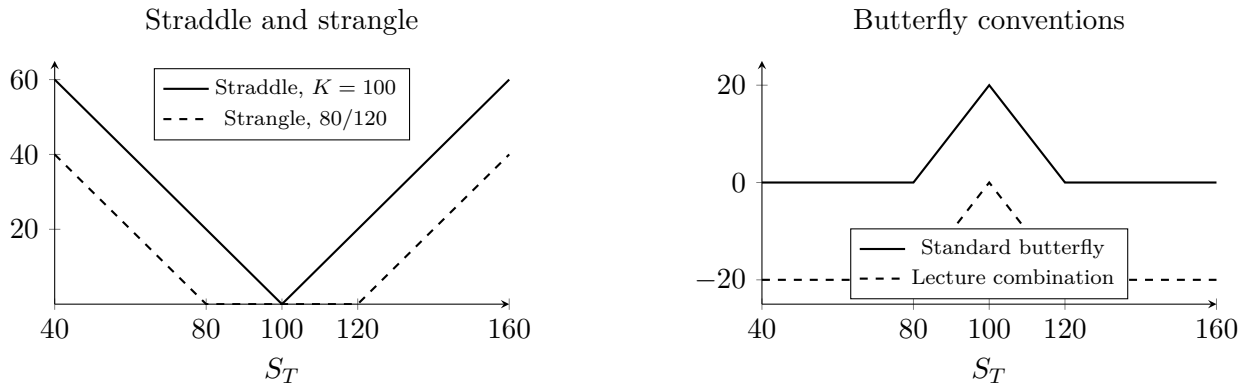


Figure 1.3: Exact payoff diagrams. The two butterfly conventions differ by a constant terminal payment; their consistently financed profits agree.

1.9.8 Reference table of terminal payoffs

Strategy	Holdings	Terminal payoff
Synthetic long forward	Long call, short put at K	$s - K$
Protective put	Long share, long put	$\max\{s, K\}$
Covered call	Long share, short call	$\min\{s, K\}$
Cap on short stock	Short share, long call	$-\min\{s, K\}$
Covered put	Short share, short put	$-\max\{s, K\}$
Bull call spread	Long call K_L, short call K_U	$(s - K_L)^+ - (s - K_U)^+$
Bear put spread	Long put K_U, short put K_L	$(K_U - s)^+ - (K_L - s)^+$
Collar	Long share, put K_L, short call K_U	$\min\{\max\{s, K_L\}, K_U\}$
Straddle	Long call and put at K	$\lvert s - K\rvert$
Strangle	Long put K_L, long call K_U	$(K_L - s)^+ + (s - K_U)^+$
Symmetric butterfly	Calls at K_L, K_U, short two at K_M	Tent payoff $B(s)$

1.9.9 Extension: replication of piecewise-linear payoffs

The strategies above are instances of one general construction. This is an explanatory extension beyond the lecture's individual examples.

Theorem 1.14 (Call-basis representation)**.** *Let $h : [0, \infty) \to \mathbb{R}$ be continuous and piecewise linear, with breakpoints $0 < K_1 < \cdots < K_m$. Let its slopes on the consecutive intervals be $b_0, b_1, \ldots, b_m$, and let $a = h(0)$. Then*

$$h(s) = a + b_0 s + \sum_{j=1}^{m} (b_j - b_{j-1})(s - K_j)^+.$$

If the required calls and prepaid share are traded, its time-t price is

$$aD + b_0 A + \sum_{j=1}^{m} (b_j - b_{j-1}) c(K_j, t, T).$$

Proof. Call the right side $g(s)$. At zero, $g(0) = a = h(0)$. On the first interval $[0, K_1)$, every positive-part term vanishes, so g has slope b_0. On (K_j, K_{j+1}), exactly the first j call terms have slope one, and therefore

$$\begin{aligned} g'(s) &= b_0 + \sum_{i=1}^{j} (b_i - b_{i-1}) \\ &= b_0 + (b_1 - b_0) + (b_2 - b_1) + \cdots + (b_j - b_{j-1}) \\ &= b_j. \end{aligned}$$

The same calculation works beyond the last breakpoint. Thus g and h have the same initial value, the same slope on every segment, and are continuous at the same breakpoints. They coincide everywhere. The formula describes a portfolio of a bond, prepaid stock, and calls, so the price follows from replication. □

Remark 1.35. A positive slope jump requires a long call; a negative slope jump requires a short call. This gives a systematic way to design a payoff rather than memorizing each named strategy.

1.10 What model-free pricing determines, and what it leaves open

Source coverage: lecture slides 13, 55, 107–108; additional worked example below..

1.10.1 Replicable versus nonreplicable payoffs

Definition 1.33 (Replicable terminal claim)**.** A terminal claim H is replicable if an admissible self-financing strategy has terminal wealth H in every relevant state. Its no-arbitrage price equals the setup cost of that strategy.

A static portfolio of a bank account and a non-dividend stock pays $a + bS_T$, an affine function of the terminal stock price. A call pays $(S_T - K)^+$, which changes slope at K. If possible terminal stock values lie on both sides of K in a sufficiently rich state space, a single affine function cannot generally match this kink.

Remark 1.36 (The scope of nonreplicability)*.* The lecture's statement that an option cannot be replicated by stock and bond is a statement about the general model-free setting. In a two-state one-period market, two independent traded payoffs can replicate every terminal claim. In some continuous-time models, dynamic stock-and-bond trading also replicates calls. These conclusions require structure not yet assumed in Part I.

1.10.2 A complete three-state example of price nonuniqueness

Example 1.14 (Different arbitrage-free call prices with the same stock and bond)**.** Consider a one-period market with $r = 0$, $S_0 = 100$, and three possible stock values

$$S_T \in \{80, 100, 120\}.$$

Assume each state has positive physical probability. A call with strike 100 pays

$$H = (0, 0, 20)$$

across the three states. Suppose a static stock-and-cash portfolio replicated it. Then its coefficients a, b would satisfy

$$\begin{aligned} a + 80b &= 0, \\ a + 100b &= 0, \\ a + 120b &= 20. \end{aligned}$$

Subtracting the first equation from the second gives $20b = 0$, so $b = 0$ and $a = 0$. The third equation would then read $0 = 20$, a contradiction. The call is not replicable with the existing assets.

For any $q \in (0, 1/2)$, define strictly positive state-price weights

$$(q_1, q_2, q_3) = (q, 1 - 2q, q).$$

These weights price the bond and stock correctly:

$$\begin{aligned} q_1 + q_2 + q_3 &= q + (1 - 2q) + q = 1, \\ 80q_1 + 100q_2 + 120q_3 &= 80q + 100(1 - 2q) + 120q \\ &= 100. \end{aligned}$$

Price every traded payoff by its weighted sum. The call then has price

$$c_0 = 0q + 0(1-2q) + 20q = 20q.$$

This pricing system is arbitrage-free. Indeed, if a portfolio has state payoffs $z_1, z_2, z_3 \geq 0$ and at least one is positive, its cost is

$$qz_1 + (1-2q)z_2 + qz_3 > 0,$$

so it cannot be a zero-cost arbitrage. If all payoffs are zero, its cost is zero, so it cannot produce a strictly positive initial receipt either.

For instance, $q = 0.2$ gives $c_0 = 4$ and $q = 0.4$ gives $c_0 = 8$. Both prices coexist with the same $S_0 = 100$, zero interest rate, and the same set of terminal stock values, in two different arbitrage-free price systems. No-arbitrage alone does not choose between them.

Remark 1.37 (State prices are not physical probabilities here). The weights were introduced as positive linear pricing coefficients. They happen to sum to one because $r = 0$. We have not asserted that they are the actual probabilities of the stock outcomes, or that physical expected payoff is the correct pricing rule.

1.10.3 A compact formula reference

Let $\Delta = T - t$, $D = e^{-r\Delta}$, and $A = S_t e^{-\delta\Delta}$ for a continuous dividend yield. Then

$$\begin{aligned} f_t(K,T) &= A - KD, & F(t,T) &= \frac{A}{D} = S_t e^{(r-\delta)\Delta}, \\ c_t - p_t &= A - KD, & c_t + KD &= p_t + A, \\ (A-KD)^+ \leq c_t &\leq A, & (KD-A)^+ \leq p_t &\leq KD. \end{aligned}$$

For known discrete dividends, replace A by $S_t - \sum_i d_i e^{-r(t_i - t)}$.

If $r \geq 0$ and $\delta \geq 0$,

$$\begin{aligned} \max\{c_t, (S_t - K)^+\} \leq C_t \leq S_t, \\ \max\{p_t, (K - S_t)^+\} \leq P_t \leq K. \end{aligned}$$

If additionally $\delta = 0$,

$$\begin{aligned} C_t &= c_t, \\ S_t - K \leq C_t - P_t &\leq S_t - KD, \\ 0 \leq P_t - p_t &\leq K(1-D). \end{aligned}$$

For $K_1 < K_2$, European strike differences satisfy

$$\begin{aligned} 0 \leq c(K_1) - c(K_2) \leq D(K_2 - K_1), \\ 0 \leq p(K_2) - p(K_1) \leq D(K_2 - K_1). \end{aligned}$$

For American strike differences with $r \geq 0$, replace D on the right by 1. All four strike-price functions are convex. American values are nondecreasing in maturity; European calls inherit this property when there are no dividends and $r \geq 0$.

1.10.4 Assumption checklist for proofs

Result or construction	Essential qualification
Law of one price	Match all remaining cash flows, not only ex-dividend prices.
Discrete-dividend forward formula	Dividend amounts and dates are known, with delivery timing specified.
Continuous-yield prepaid stock	Dividends are reinvested; $e^{-\delta\Delta}$ is a share count.
American put bound $P_t \leq K$	Nonnegative financing rate.
American call equals European call	No dividends and nonnegative financing rate.
European 1-Lipschitz strike bound	Nonnegative rate; otherwise use D-Lipschitz.
American convexity and strike bounds	A covering response at every short exercise time.
European maturity comparison	No unconditional ordering; use stated special cases.
Payoff-based trading profit	Subtract the financed initial cost and include intermediate cash flows.
Futures versus forwards	Settlement timing differs; quote equality needs financing assumptions.

1.11 Worked diagnostic exercises

These exercises consolidate the constructions, including the proofs left open in the lecture. They introduce no additional pricing model.

1.11.1 A put quote below its lower bound

Example 1.15. Assume no dividends, $S_0 = 90$, $K = 100$, $r = 0$, and a European put quote $p_0 = 7$. The lower bound is $(100 - 90)^+ = 10$, so the put is underpriced.

Buy one put, buy one share, and borrow 100. Setup produces

$$-7 - 90 + 100 = 3.$$

The terminal payoff of those positions is

$$(100 - S_T)^+ + S_T - 100 = (S_T - 100)^+ \geq 0.$$

Keep the three-dollar receipt in the bank. The final wealth is at least three dollars in every state. This explicitly converts a lower-bound violation into arbitrage.

1.11.2 A vertical spread that is too expensive

Example 1.16. Let $K_1 = 100$, $K_2 = 110$, $r = 0$, and suppose European call quotes are $c(100) = 15$ and $c(110) = 3$. Their difference is 12, exceeding the maximum terminal spread payoff 10.

Sell the lower-strike call, buy the higher-strike call, and put 10 in the bank. The initial cash flow is

$$15 - 3 - 10 = 2.$$

The terminal payoff is

$$10 - (S_T - 100)^+ + (S_T - 110)^+ = 10 - H_{\mathrm{BC}}(S_T) \geq 0.$$

Banking the remaining two dollars yields guaranteed positive wealth. For general r, the bond needed for a European spread costs $10e^{-r(T-t)}$, whereas a hedge against American exercise at any time can use 10 dollars initially when $r \geq 0$.

1.11.3 Designing a capped loss-protection payoff

Example 1.17. Construct a terminal payoff that equals 90 below stock price 90, increases one-for-one from 90 to 110, and stays at 110 thereafter. This is

$$h(s) = \begin{cases} 90, & s \le 90, \\ s, & 90 < s < 110, \\ 110, & s \ge 110. \end{cases}$$

The initial slope is zero, the slope jumps up by one at 90 and down by one at 110. Hence the call-basis representation is

$$h(s) = 90 + (s - 90)^+ - (s - 110)^+.$$

Buy a bond paying 90, buy a call at 90, and sell a call at 110. Its price is

$$90D + c(90) - c(110).$$

Equivalently, buy prepaid stock, buy a put at 90, and sell a call at 110. Parity verifies the same cost:

$$A + p(90) - c(110) = 90D + c(90) - c(110).$$

1.11.4 Recovering a missing option quote

Example 1.18. Suppose $A = 98$, $KD = 100$, and a European call is priced at 6. Put-call parity requires

$$\begin{aligned} c - p &= A - KD = -2, \\ 6 - p &= -2, \\ p &= 8. \end{aligned}$$

The bound $p \ge KD - A = 2$ is satisfied, but it is parity with the traded call, rather than the bound alone, that identifies the put price. Without the call quote, many put prices could be consistent with the individual bounds.

Chapter 2

Discrete-Time Models

2.1 From model-free restrictions to a stock-price model

Source coverage: lecture slides 1–5.. Chapter 1 compared attainable cash flows without specifying a law for the stock. This gave exact forward prices and relations between option prices, but generally did not identify an individual option price. We now specify which stock-price movements are possible and when trading is permitted. The resulting market can admit dynamic replication of nonlinear claims.

The chapter proceeds from one period and two states to many periods, early exercise, dividends, and path-dependent claims. It ends with a general finite-state market, where replication becomes a linear algebra problem. All examples in the supplied 136-slide lecture are expanded below, and the omitted converse and completeness proofs are supplied.

2.1.1 Time indices and standing assumptions

Let $N \geq 1$ be the number of periods, let $h > 0$ be their common length, and write $t_n = nh$ and $T = Nh$. To simplify formulas, use

$$S_n := S_{t_n}, \qquad B_n = e^{rnh} = R^n, \qquad R := e^{rh} > 0.$$

Thus S_1 means the stock price at calendar time h, not necessarily at one year. The gross interest factor is R; the one-period discount factor is R^{-1}. Unless explicitly discussing dividends, stocks pay none. The assumptions about unrestricted fractional trading, short sales, and common borrowing/lending rates from Chapter 1 remain in force.

Every state in the finite sample space has strictly positive physical probability. All claims and strategy holdings are finite real numbers. Hence all random variables appearing in finite-tree conditional expectations are integrable, and there is no need for an additional limiting or uniform-integrability argument.

2.1.2 Physical probabilities versus pricing probabilities

Definition 2.1 (Physical measure)**.** The physical measure $\mathbb{P}$ describes the assumed actual distribution of future stock-price movements. Its role is to describe possible events and their likelihoods under the statistical model.

Definition 2.2 (Equivalent probability measures)**.** Two probability measures $\mathbb{P}$ and $\mathbb{Q}$ are equivalent, written $\mathbb{Q} \sim \mathbb{P}$, if they have the same null events. On a finite sample space with $\mathbb{P}(\{\omega\}) > 0$ for every state, equivalence means $\mathbb{Q}(\{\omega\}) > 0$ for every state as well.

Strict positivity will be essential: a pricing measure that assigns zero probability to a possible profitable state can overlook an arbitrage. Calling $\mathbb{Q}$ a pricing measure does not assert that investors are literally risk-neutral or that $\mathbb{Q}$ gives empirically correct forecasts.

2.2 The one-period binomial market

Source coverage: lecture slides 6–18..

2.2.1 Two possible stock prices and one bank account

Definition 2.3 (One-period binomial model)**.** Let $S_0 > 0$ and $0 < d < u$. There are two states $\Omega = \{U, D\}$, with

$$\begin{aligned} S_1(U) &= uS_0, & \mathbb{P}(U) &= p \in (0,1), \\ S_1(D) &= dS_0, & \mathbb{P}(D) &= 1 - p. \end{aligned}$$

The bank account has prices $B_0 = 1$ and $B_1 = R = e^{rh}$.

The names "up" and "down" mean higher and lower relative outcomes. One need not assume $d < 1 < u$; the economically relevant comparison for arbitrage is with R.

Definition 2.4 (One-period portfolio)**.** A portfolio $(\Delta, b) \in \mathbb{R}^2$ holds Δ shares and invests b dollars in the bank account at time zero. Its values are

$$\begin{aligned} V_0 &= \Delta S_0 + b, \\ V_1(U) &= \Delta u S_0 + bR, \\ V_1(D) &= \Delta d S_0 + bR. \end{aligned}$$

A negative b is a loan; a negative Δ is a short stock position.

Definition 2.5 (Arbitrage in the binomial model)**.** A zero-cost arbitrage has $V_0 = 0$ and terminal payoffs satisfying

$$V_1(U) \geq 0, \qquad V_1(D) \geq 0,$$

with at least one of them strictly positive. An initial receipt can instead be invested in the bank account to produce a zero-cost strategy.

Because both states have positive physical probability, "positive in at least one state" is equivalent to "positive with positive probability".

2.2.2 Both directions of the no-arbitrage condition

Theorem 2.1 (Exact one-period no-arbitrage condition)**.** *The one-period binomial market is arbitrage-free if and only if*

$$\boxed{d < R < u.}$$

Proof. **Necessity, upper inequality.** If $R \geq u$, short one share and lend the proceeds S_0. The initial cost is zero, while

$$\begin{aligned} V_1(U) &= -uS_0 + S_0R = S_0(R-u) \geq 0, \\ V_1(D) &= -dS_0 + S_0R = S_0(R-d) > 0. \end{aligned}$$

The second inequality is strict because $R \geq u > d$. This is arbitrage, including the boundary case $R = u$.

Necessity, lower inequality. If $R \leq d$, buy one share and borrow S_0. Then

$$\begin{aligned} V_0 &= S_0 - S_0 = 0, \\ V_1(U) &= uS_0 - S_0R = S_0(u-R) > 0, \\ V_1(D) &= dS_0 - S_0R = S_0(d-R) \geq 0. \end{aligned}$$

This is arbitrage, including $R = d$. Therefore no-arbitrage requires $d < R < u$.

Sufficiency. Assume $d < R < u$ and set

$$q = \frac{R-d}{u-d}, \qquad 1-q = \frac{u-R}{u-d}.$$

Both weights are strictly positive and sum to one. For every portfolio,

$$\begin{aligned} R^{-1}(qV_1(U) + (1-q)V_1(D)) &= R^{-1}[\Delta S_0(qu + (1-q)d) + bR] \\ &= R^{-1}(\Delta S_0 R + bR) \\ &= \Delta S_0 + b = V_0. \end{aligned}$$

If both terminal values are nonnegative and at least one is strictly positive, the weighted sum is strictly positive, so $V_0 > 0$. Such a payoff cannot be obtained at zero or negative cost. Also, if both payoffs vanish, the displayed identity gives $V_0 = 0$, ruling out a negative-cost zero-payoff trade. Thus there is no arbitrage. □

2.2.3 Constructing the equivalent martingale measure

Lemma 2.1 (A strictly interior convex combination)**.** *For $u > d$, the equation*

$$R = qu + (1-q)d$$

has the unique solution $q = (R-d)/(u-d)$. This solution belongs to $(0,1)$ exactly when $d < R < u$.

Proof. Rearrange as

$$\begin{aligned} R &= d + q(u-d), \\ q(u-d) &= R - d, \\ q &= \frac{R-d}{u-d}. \end{aligned}$$

Because $u - d > 0$, $q > 0$ is equivalent to $R > d$, and $q < 1$ is equivalent to $R - d < u - d$, or $R < u$. □

Definition 2.6 (One-period equivalent martingale measure)**.** An equivalent martingale measure is a probability measure $\mathbb{Q} \sim \mathbb{P}$ satisfying

$$S_0 = R^{-1}\mathbb{E}^{\mathbb{Q}}[S_1].$$

In this model, $\mathbb{Q}(U) = q$ and $\mathbb{Q}(D) = 1 - q$ with $q = (R-d)/(u-d)$.

Theorem 2.2 (First fundamental theorem in the binomial model)**.** *The one-period binomial market is arbitrage-free if and only if an equivalent martingale measure exists. When it exists, it is unique.*

Proof. No-arbitrage is equivalent to $d < R < u$. The preceding lemma is equivalent to existence of strictly positive weights satisfying

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[S_1] &= q(uS_0) + (1-q)(dS_0) \\ &= S_0 R. \end{aligned}$$

The equation for q has a unique solution. □

Remark 2.1 (Why equivalence cannot be omitted). At $R = u$, the weight $q = 1$ gives $S_0 = R^{-1}\mathbb{E}^{\mathbb{Q}}[S_1]$, but the short-stock arbitrage above earns a strictly positive gain in state D, which this measure ignores. The theorem requires a strictly positive weight on every physically possible state, not merely a nonnegative probability measure.

2.3 Replication, completeness, and one-period valuation

Source coverage: lecture slides 19–27..

2.3.1 Solving the replication equations

Definition 2.7 (Attainable claim and complete market)**.** A claim X is attainable if some admissible trading strategy delivers X in every state. A market is complete, in the convention of these notes and the lecture, if it is arbitrage-free and every claim measurable with respect to its specified terminal information is attainable.

Write $X_U = X(U)$ and $X_D = X(D)$. On a two-state space, every claim is determined by these two numbers; since $uS_0 \neq dS_0$, it can also be written as $\Phi(S_1)$.

Theorem 2.3 (Unique one-period replicating portfolio)**.** *Every two-state claim has a unique stock-and-bank replicating portfolio:*

$$\boxed{\Delta = \frac{X_U - X_D}{S_0(u-d)}, \qquad b = \frac{uX_D - dX_U}{R(u-d)}.}$$

Consequently, an arbitrage-free one-period binomial market is complete.

Proof. The required equations are

$$\begin{aligned} \Delta u S_0 + bR &= X_U, \\ \Delta d S_0 + bR &= X_D. \end{aligned}$$

Subtract the second from the first:

$$\begin{aligned} \Delta S_0(u-d) &= X_U - X_D, \\ \Delta &= \frac{X_U - X_D}{S_0(u-d)}. \end{aligned}$$

Substitute into the down-state equation:

$$\begin{aligned} bR &= X_D - \Delta d S_0 \\ &= X_D - \frac{d(X_U - X_D)}{u-d} \\ &= \frac{(u-d)X_D - dX_U + dX_D}{u-d} \\ &= \frac{uX_D - dX_U}{u-d}, \\ b &= \frac{uX_D - dX_U}{R(u-d)}. \end{aligned}$$

Every denominator is nonzero. Equivalently, the payoff matrix

$$\begin{pmatrix} uS_0 & R \\ dS_0 & R \end{pmatrix}$$

has determinant $RS_0(u-d) > 0$, so the solution exists and is unique. □

Remark 2.2 (Delta is a secant hedge ratio). The number Δ is the change in the claim's two possible values divided by the change in the stock's two possible values. It is an exact finite-state hedge ratio. Identifying it with a derivative $\partial V/\partial S$ requires an additional smooth or limiting framework.

2.3.2 Deriving discounted risk-neutral expectation from replication

Theorem 2.4 (One-period risk-neutral valuation)**.** *The unique arbitrage-free price of X is*

$$\boxed{\pi_0(X) = R^{-1}\big(qX_U + (1-q)X_D\big) = e^{-rh}\mathbb{E}^{\mathbb{Q}}[X].}$$

Proof. The claim must have the price of its replicating portfolio. Expanding the cost gives

$$\begin{aligned}
\pi_0(X) &= \Delta S_0 + b \\
&= \frac{X_U - X_D}{u-d} + \frac{uX_D - dX_U}{R(u-d)} \\
&= \frac{R(X_U - X_D) + uX_D - dX_U}{R(u-d)} \\
&= R^{-1}\left[\frac{R-d}{u-d}X_U + \frac{u-R}{u-d}X_D\right] \\
&= R^{-1}\big(qX_U + (1-q)X_D\big).
\end{aligned}$$

If the claim were dearer, sell it and buy its hedge; if cheaper, buy it and sell the hedge. Bank the initial receipt. Terminal obligations cancel exactly, so either discrepancy produces arbitrage. □

Remark 2.3 (Why p disappears)*.* The hedge matches the payoff in each state separately. Changing p while keeping both states possible does not change those statewise equations. It can change expected trading profit under $\mathbb{P}$, but not the cost of exact replication. This conclusion holds conditional on the specified S_0, u, d, r; changing statistical assumptions might lead one to recalibrate those model inputs.

Remark 2.4 (Discounting physical expectation is generally wrong)*.* In general,

$$R^{-1}\big(pX_U + (1-p)X_D\big) \neq \pi_0(X).$$

The bank rate discounts the replicating portfolio's *pricing-measure* expectation. No argument so far says that the physical expected return on a risky asset is the bank rate.

2.3.3 The lecture's six-month call, including the hedge

Example 2.1. Take $S_0 = 100$, $u = 1.1$, $d = 0.9$, $h = 0.5$, $r = 0.05$, and $K = 105$. Then

$$\begin{aligned}
R &= e^{0.025} \approx 1.02531512, \\
q &= \frac{e^{0.025} - 0.9}{0.2} \approx 0.62657560, \\
X_U &= (110-105)^+ = 5, \qquad X_D = (90-105)^+ = 0.
\end{aligned}$$

The no-arbitrage condition $0.9 < R < 1.1$ holds. Therefore

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[X] &= 5q \approx 3.13287801, \\
c_0 &= e^{-0.025}(5q) \approx 3.05552698.
\end{aligned}$$

The exact hedge is

$$\begin{aligned}\Delta &= \frac{5-0}{110-90} = \frac{1}{4},\\ b &= \frac{1.1(0)-0.9(5)}{0.2R} = -\frac{22.5}{R} \approx -21.94447302,\\ \Delta S_0 + b &= 25 - \frac{22.5}{R} \approx 3.05552698.\end{aligned}$$

At maturity,

$$\begin{aligned}V_1(U) &= \tfrac{1}{4}(110) - 22.5 = 5,\\ V_1(D) &= \tfrac{1}{4}(90) - 22.5 = 0.\end{aligned}$$

Thus buying a quarter share and borrowing the present value of 22.5 replicates the call exactly.

2.4 The multiperiod tree and self-financing trading

Source coverage: lecture slides 28–41, 47–49..

2.4.1 Paths, recombination, and information

Definition 2.8 (Homogeneous binomial tree)**.** Let $\Omega = \{U, D\}^N$. Under $\mathbb{P}$, let $Z_1, \dots, Z_N$ be independent with

$$\mathbb{P}(Z_n = u) = p, \qquad \mathbb{P}(Z_n = d) = 1 - p.$$

Define $S_{n+1} = S_n Z_{n+1}$ and let $\mathcal{F}_n = \sigma(Z_1, \dots, Z_n)$, with trivial $\mathcal{F}_0$.

Proposition 2.1 (Stock values at a level)**.** *If J_n is the number of up moves in the first n steps, then*

$$S_n = S_0 u^{J_n} d^{n-J_n}.$$

There are 2^n histories of length n, but only $n+1$ distinct stock prices at that time.

Proof. Iterate the recursion to obtain $S_n = S_0 \prod_{i=1}^n Z_i$. Multiplication commutes, so only the counts of the two factors matter. Since $u/d > 1$, the prices $S_0 d^n (u/d)^j$, $j = 0, \dots, n$, are distinct. □

Remark 2.5 (A recombining stock is not the same as recombining information). Histories UD and DU both lead to $S_2 = S_0 ud$, but the observed paths differ. A path-dependent claim can have different values after these histories. Conditioning on $\mathcal{F}_n$ is therefore more informative than conditioning only on S_n.

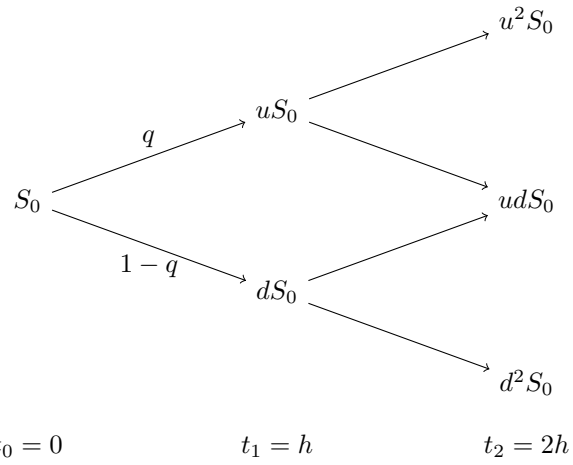


Figure 2.1: A two-period recombining stock tree under the pricing measure. The middle terminal price corresponds to two distinct histories.

2.4.2 Predictable holdings and the cash-account convention

Definition 2.9 (Trading strategy)**.** At time t_n, choose Δ_{n+1} shares and b_{n+1} dollars in the bank to hold over $(t_n, t_{n+1}]$. Both choices must be $\mathcal{F}_n$-measurable. Thus the holding used to earn the next increment is chosen before that increment is observed.

Here b_{n+1} is a *dollar amount at time* t_n, not a fixed number of units of B. It becomes Rb_{n+1} at the next date. This convention matches the multiperiod cash variable in the lecture and avoids confusing it with bank-account units.

Definition 2.10 (Self-financing condition)**.** The initial cost is $V_0 = \Delta_1 S_0 + b_1$. For $n = 1, \ldots, N-1$, self-financing requires

$$\boxed{\Delta_n S_n + Rb_n = \Delta_{n+1} S_n + b_{n+1} = V_n.}$$

At the final time, $V_N = \Delta_N S_N + Rb_N$.

The left side is wealth arriving from the previous holdings. The right side is the cost of the new holdings. Equality means that rebalancing needs no outside contribution and makes no withdrawal.

Lemma 2.2 (Wealth and discounted-gains recursions)**.** *A self-financing strategy satisfies*

$$V_{n+1} = RV_n + \Delta_{n+1}(S_{n+1} - RS_n),$$
$$\frac{V_{n+1}}{R^{n+1}} - \frac{V_n}{R^n} = \Delta_{n+1}\left(\frac{S_{n+1}}{R^{n+1}} - \frac{S_n}{R^n}\right).$$

Consequently,

$$\frac{V_n}{R^n} = V_0 + \sum_{i=0}^{n-1} \Delta_{i+1}\left(\frac{S_{i+1}}{R^{i+1}} - \frac{S_i}{R^i}\right).$$

Proof. Since $b_{n+1} = V_n - \Delta_{n+1} S_n$,

$$\begin{aligned} V_{n+1} &= \Delta_{n+1} S_{n+1} + Rb_{n+1} \\ &= \Delta_{n+1} S_{n+1} + R(V_n - \Delta_{n+1} S_n) \\ &= RV_n + \Delta_{n+1}(S_{n+1} - RS_n). \end{aligned}$$

Divide by R^{n+1} and subtract V_n/R^n. Summing the resulting increments telescopes to the last formula. □

Remark 2.6 (Connection with predictable martingale transforms). The discounted-gains formula is exactly a discrete stochastic integral: predictable holdings multiply increments of the discounted stock. This connects the financial self-financing condition directly to the martingale-transform results in probability theory.

2.4.3 Constructing the path-space pricing measure

Theorem 2.5 (Unique equivalent martingale measure on the tree)**.** *Assume $d < R < u$. Define $q = (R-d)/(u-d)$ and, for a full path ω with j up moves, set*

$$\mathbb{Q}(\{\omega\}) = q^j(1-q)^{N-j}.$$

This is the unique equivalent measure for which $(S_n/R^n)_{n=0}^N$ is a martingale with respect to $(\mathcal{F}_n)$.

Proof. All path probabilities are positive, and their sum is $(q+(1-q))^N = 1$. Under this product measure the factors remain independent, now with up probability q. Hence

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[S_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}^{\mathbb{Q}}[S_n Z_{n+1} \mid \mathcal{F}_n] \\ &= S_n \mathbb{E}^{\mathbb{Q}}[Z_{n+1} \mid \mathcal{F}_n] \\ &= S_n(qu + (1-q)d) \\ &= RS_n.\end{aligned}$$

Dividing by R^{n+1} proves the martingale property.

For uniqueness, let any equivalent martingale measure give conditional up probability $q_n(\omega)$ after a particular history. The martingale condition requires

$$\begin{aligned}S_n(q_n u + (1-q_n)d) &= RS_n, \\ q_n &= \frac{R-d}{u-d} = q,\end{aligned}$$

since $S_n > 0$. Thus every node has the same forced conditional probabilities. Multiplying along a path gives the displayed product law, proving uniqueness. □

2.4.4 Discounted self-financing wealth is a martingale

Proposition 2.2 (Risk-neutral growth of a trading portfolio)**.** *For every self-financing strategy,*

$$\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n] = RV_n.$$

Thus (V_n/R^n) *is a martingale.*

Proof. Predictability lets us take the chosen holdings outside the conditional expectation:

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}^{\mathbb{Q}}[\Delta_{n+1}S_{n+1} + Rb_{n+1} \mid \mathcal{F}_n] \\
&= \Delta_{n+1}\mathbb{E}^{\mathbb{Q}}[S_{n+1} \mid \mathcal{F}_n] + Rb_{n+1} \\
&= \Delta_{n+1}RS_n + Rb_{n+1} \\
&= R(\Delta_{n+1}S_n + b_{n+1}) \\
&= RV_n.
\end{aligned}$$

The last equality is self-financing. Every expectation is finite because the tree and all holdings are finite. □

Theorem 2.6 (No-arbitrage on the multiperiod tree)**.** *The homogeneous multiperiod binomial market is arbitrage-free if and only if* $d < R < u$.

Proof. If the inequality fails, execute the corresponding one-period arbitrage over the first step and put its nonnegative proceeds in the bank for the remaining steps. The terminal payoff stays nonnegative and is strictly positive on a possible event.

If the inequality holds, the preceding martingale result gives

$$V_0 = \mathbb{E}^{\mathbb{Q}}[V_N/R^N].$$

A nonnegative terminal payoff that is positive on a physical event of positive probability is also positive on an event of positive $\mathbb{Q}$ probability, by equivalence. Its discounted expectation is therefore strictly positive. It cannot come from zero or negative initial wealth. This proves sufficiency. □

2.5 Dynamic completeness and multiperiod valuation

Source coverage: lecture slides 42–53..

2.5.1 A complete backward-induction replication proof

Theorem 2.7 (Every finite-tree claim is dynamically attainable)**.** *Let X be any $\mathcal{F}_N$-measurable claim, including a claim depending on the entire stock path. In the arbitrage-free binomial model, define values on histories by*

$$\begin{aligned} v_N &= X, \\ v_n &= R^{-1}(qv_{n+1}^U + (1-q)v_{n+1}^D), \qquad n = N-1, \dots, 0. \end{aligned}$$

Here v_{n+1}^U and v_{n+1}^D are the values at the two child histories of the current history. The claim is replicated by the predictable holdings

$$\begin{aligned} \Delta_{n+1} &= \frac{v_{n+1}^U - v_{n+1}^D}{S_n(u-d)}, \\ b_{n+1} &= \frac{uv_{n+1}^D - dv_{n+1}^U}{R(u-d)}. \end{aligned}$$

The wealth at each node is v_n.

Proof. **Terminal condition.** At time N, assign the required value $v_N = X$ at each full history.

One backward step. Suppose values have already been assigned at every history of length $n+1$. At a history of length n, the one-period replication theorem gives the unique holdings whose two possible next values are those assigned child values. Explicitly,

$$\begin{aligned} \Delta_{n+1} uS_n + Rb_{n+1} &= v_{n+1}^U, \\ \Delta_{n+1} dS_n + Rb_{n+1} &= v_{n+1}^D, \\ \Delta_{n+1} S_n + b_{n+1} &= v_n. \end{aligned}$$

The holdings depend only on the current history, so they are $\mathcal{F}_n$-measurable.

Joining the steps. Start with v_0 and buy the root holdings. After one step the portfolio is worth the assigned v_1 at whichever child is reached. By construction, v_1 is exactly the cost of the next local hedge. Therefore rebalancing satisfies

$$\Delta_1 S_1 + Rb_1 = v_1 = \Delta_2 S_1 + b_2.$$

Inductively, arriving at any history at level n gives wealth v_n, precisely the cost of its chosen next holdings. No external money is needed. After the last step, the wealth is $v_N = X$. This constructs a self-financing replication for every terminal claim. □

Remark 2.7 (Why dynamic replication succeeds). At a given node only two next-state values need to be matched, so two traded instruments suffice locally. A single buy-and-hold stock-and-bank portfolio has only two constant coefficients and generally cannot fit all terminal payoffs once there are more than two terminal states. Rebalancing creates enough history-dependent degrees of freedom.

Remark 2.8 (The information space matters). Completeness refers to claims measurable with respect to the stock-generated finite tree. If the terminal information also contained an independent untraded random variable, claims depending on that variable need not be attainable. Two branches per stock node do not by themselves insure unrelated sources of uncertainty.

2.5.2 Pricing at every node

Proposition 2.3 (Nodewise law of one price)**.** *The arbitrage-free price process of X equals the replicating wealth process:*

$$\pi_n(X) = v_n \qquad (n = 0, \dots, N).$$

In particular, the price process is adapted.

Proof. At a reached history, the future hedge matches every remaining payment of the claim. If their current prices differed, buy the cheaper and sell the dearer, then invest the receipt. The future matched payoffs cancel. Each history has positive physical probability, so implementing this zero-cost trade only when that history is reached also creates an arbitrage from the initial date. Hence prices agree at every node. The constructed v_n is a function of the history through n, proving adaptedness. □

Theorem 2.8 (Conditional risk-neutral valuation)**.** *For every terminal claim X,*

$$\boxed{\pi_n(X) = R^{-(N-n)}\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_n].}$$

Equivalently, at calendar time t_n,

$$\pi_{t_n}(X) = e^{-r(T-t_n)}\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_n].$$

Proof. At $n = N$, X is $\mathcal{F}_N$-measurable, so

$$\pi_N(X) = X = \mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_N].$$

Assume the formula holds at $n + 1$. The local replication formula and the tower property give

$$\begin{aligned}
\pi_n(X) &= R^{-1}\mathbb{E}^{\mathbb{Q}}[\pi_{n+1}(X) \mid \mathcal{F}_n] \\
&= R^{-1}\mathbb{E}^{\mathbb{Q}}\Big[R^{-(N-n-1)}\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\Big] \\
&= R^{-(N-n)}\mathbb{E}^{\mathbb{Q}}\Big[\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\Big] \\
&= R^{-(N-n)}\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_n].
\end{aligned}$$

The last step uses $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$. Backward induction proves the identity for every n. □

Corollary 2.1 (Discounted derivative prices are martingales)**.** *The process $\pi_n(X)/R^n$ is a $\mathbb{Q}$-martingale, and $\pi_0(X) = R^{-N}\mathbb{E}^{\mathbb{Q}}[X]$.*

Proof. Divide the conditional valuation formula by R^n to obtain $R^{-N}\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_n]$. The tower property proves the martingale identity. At zero, the conditioning is trivial. □

2.6 European options: recursion, terminal sums, and explicit hedges

Source coverage: lecture slides 54–71..

2.6.1 When the current stock price is a sufficient state

For a claim $X = \Phi(S_N)$ in the homogeneous tree, future factors are independent of the past under $\mathbb{Q}$. Thus the current stock price and the number of remaining periods determine the conditional payoff distribution. Write $v_{n,j}$ for the value at time n after j up moves:

$$\begin{aligned} s_{n,j} &= S_0 u^j d^{n-j}, \\ v_{N,j} &= \Phi(s_{N,j}), \\ v_{n,j} &= R^{-1}(q v_{n+1,j+1} + (1-q) v_{n+1,j}). \end{aligned}$$

The corresponding next-period holdings are

$$\begin{aligned} \Delta_{n+1,j} &= \frac{v_{n+1,j+1} - v_{n+1,j}}{s_{n,j}(u-d)}, \\ b_{n+1,j} &= \frac{u v_{n+1,j} - d v_{n+1,j+1}}{R(u-d)}. \end{aligned}$$

For a path-dependent claim, these formulas still work at a history node, but j alone need not identify that node's derivative value.

2.6.2 The two-period European put from the lecture

Example 2.2 (Tree and terminal payoffs)**.** Let

$$S_0 = 150, \quad K = 160, \quad T = 0.5, \quad N = 2, \quad h = 0.25, \quad u = 1.3, \quad d = 0.7, \quad r = 0.06.$$

Then

$$\begin{aligned} R &= e^{0.015} \approx 1.01511306, \\ q &= \frac{e^{0.015} - 0.7}{0.6} \approx 0.52518844, \\ 1 - q &\approx 0.47481156. \end{aligned}$$

The stock values and put payoffs are

History	Stock price	Terminal put payoff
UU	$150(1.3)^2 = 253.5$	0
UD or DU	$150(1.3)(0.7) = 136.5$	23.5
DD	$150(0.7)^2 = 73.5$	86.5

At the first up node $S_1 = 195$,

$$\begin{aligned} v_{1,U} &= R^{-1}(q \cdot 0 + (1-q)23.5) \\ &\approx 10.99194959. \end{aligned}$$

At the first down node $S_1 = 105$,

$$\begin{aligned} v_{1,D} &= R^{-1}(q 23.5 + (1-q)86.5) \\ &\approx 52.61791034. \end{aligned}$$

Finally,

$$\begin{aligned} p_0 &= R^{-1}(qv_{1,U} + (1-q)v_{1,D}) \\ &\approx 30.29853322. \end{aligned}$$

All calculations use unrounded q internally; displayed intermediate values are rounded only for readability.

Example 2.3 (The actual self-financing hedge)**.** At the root, the required stock and bank holdings are

$$\begin{aligned} \Delta_1 &= \frac{10.99194959 - 52.61791034}{195 - 105} \approx -0.46251067, \\ b_1 &= p_0 - 150\Delta_1 \approx 99.67513446. \end{aligned}$$

Thus short about 0.46251 shares and place about 99.67513 in the bank. After an up move, rebalance to

$$\begin{aligned} \Delta_2^U &= \frac{0 - 23.5}{253.5 - 136.5} = -\frac{23.5}{117} \approx -0.20085470, \\ b_2^U &= \frac{1.3(23.5) - 0.7(0)}{0.6R} \approx 50.15861626. \end{aligned}$$

After a down move, use

$$\begin{aligned} \Delta_2^D &= \frac{23.5 - 86.5}{136.5 - 73.5} = -1, \\ b_2^D &= \frac{160}{R} \approx 157.61791034. \end{aligned}$$

The rebalancing costs agree with arriving wealth:

$$\begin{aligned} 195\Delta_2^U + b_2^U &\approx 10.99194959 = v_{1,U}, \\ 105\Delta_2^D + b_2^D &= -105 + 157.61791034 = v_{1,D}. \end{aligned}$$

The root holdings arrive at these same values by their defining replication equations. At the lower node, the put will be in the money after either next move, so its remaining payoff is exactly $160 - S_2$. This explains the particularly simple hedge: short one share and lend the present value of 160.

2.6.3 The binomial terminal-sum formula

Theorem 2.9 (Direct valuation for terminal-price claims)**.** *For $X = \Phi(S_N)$,*

$$\boxed{\pi_0(X) = R^{-N} \sum_{j=0}^{N} \binom{N}{j} q^j (1-q)^{N-j} \Phi(S_0 u^j d^{N-j}).}$$

At a node with current price s and $m = N - n$ periods remaining,

$$\pi_n(X) = R^{-m} \sum_{j=0}^{m} \binom{m}{j} q^j (1-q)^{m-j} \Phi(s u^j d^{m-j}).$$

Proof. Exactly $\binom{N}{j}$ paths have j up moves. Independence gives each such path probability $q^j(1-q)^{N-j}$. All of those paths have the same terminal stock price. Therefore

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[\Phi(S_N)] &= \sum_{\omega \in \Omega} \mathbb{Q}(\{\omega\}) \Phi(S_N(\omega)) \\
&= \sum_{j=0}^{N} \sum_{\omega : J_N(\omega) = j} q^j (1-q)^{N-j} \Phi(S_0 u^j d^{N-j}) \\
&= \sum_{j=0}^{N} \binom{N}{j} q^j (1-q)^{N-j} \Phi(S_0 u^j d^{N-j}).
\end{aligned}$$

Discount by R^N. Conditional on a current node, repeat the same argument with the m independent remaining factors. □

Remark 2.9 (Correction of the general formula on slide 71)*.* For N periods, the down exponents are $N - j$, not $2 - j$. The latter is valid only in the two-period example. The general formula above also requires a payoff depending only on terminal stock price; grouping histories by the final up count is not valid for an arbitrary path-dependent payoff.

Example 2.4 (Checking the two-period put by a terminal sum)**.**

$$\begin{aligned}
p_0 &= e^{-0.03} \left[q^2(0) + 2q(1-q)(23.5) + (1-q)^2(86.5) \right] \\
&\approx 30.29853322.
\end{aligned}$$

The factor two counts the distinct histories UD and DU. This agrees with backward induction and the replicating-portfolio cost.

Remark 2.10 (Computational cost)*.* A recombining tree contains $\sum_{n=0}^{N}(n+1) = (N+1)(N+2)/2$ nodes, so full backward induction costs $O(N^2)$ arithmetic operations and can use $O(N)$ memory by keeping one level at a time. A direct terminal sum has $N+1$ terms; recursively updating the binomial weights and stock levels gives $O(N)$ arithmetic work. Numerically stable implementations may use log probabilities or recurrences centred near the binomial mode to avoid underflow. The direct sum does not provide all intermediate hedges or exercise decisions.

2.7 American exercise as a finite optimal-stopping problem

Source coverage: lecture slides 72–83..

2.7.1 Exercise on the trading grid

In this discrete model, exercise is permitted at $t_0, \ldots, t_N$. Thus "American" means American relative to the model's trading grid; viewed in continuous calendar time, these are Bermudan exercise rights. No between-date stock path has been specified.

Definition 2.11 (Exercise payoff and permitted stopping times)**.** Let $G_n \geq 0$ be the adapted exercise payoff at time n. For a vanilla put, $G_n = (K - S_n)^+$. Let $\mathcal{T}_n$ be the set of stopping times with values in $\{n, \ldots, N\}$. A holder choosing $\tau \in \mathcal{T}_n$ receives G_τ at t_τ.

A prescribed stopping rule is an attainable dated cash flow. Its value at time n is

$$\mathbb{E}^{\mathbb{Q}}[R^{-(\tau-n)} G_\tau \mid \mathcal{F}_n].$$

To see the timing precisely, first replicate the terminal claim $R^{N-\tau} G_\tau$. At the stopping date its replication value is G_τ, since both τ and G_τ are then known. Liquidating the hedge pays the exercise amount at the required date.

2.7.2 The backward recursion and optimality proof

Theorem 2.10 (American value and optimal exercise)**.** *Define*

$$\begin{aligned} A_N &= G_N, \\ H_n &= R^{-1}\mathbb{E}^{\mathbb{Q}}[A_{n+1} \mid \mathcal{F}_n], \\ A_n &= \max\{G_n, H_n\}, \qquad n = N-1, \ldots, 0. \end{aligned}$$

Then

$$\boxed{A_n = \max_{\tau \in \mathcal{T}_n} \mathbb{E}^{\mathbb{Q}}[R^{-(\tau-n)} G_\tau \mid \mathcal{F}_n].}$$

An optimal rule is

$$\tau_n^* = \inf\{k \in \{n, \ldots, N\} : A_k = G_k\}.$$

The set is nonempty because $A_N = G_N$. At a tie between continuation and exercise, either action can be optimal; the displayed rule chooses exercise.

Proof. The event that the first equality occurs by k is determined by the observations through k, so τ_n^* is a stopping time.

We prove the value identity by backward induction. At N, the only available exercise time is N, and $A_N = G_N$. Suppose it holds at $n+1$. A stopping rule at n either exercises now or waits at least one step. If it waits, conditional optimality at the next node and the tower property give

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[R^{-(\tau-n)} G_\tau \mid \mathcal{F}_n] &= R^{-1}\mathbb{E}^{\mathbb{Q}}\Big[\mathbb{E}^{\mathbb{Q}}[R^{-(\tau-n-1)} G_\tau \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\Big] \\ &\leq R^{-1}\mathbb{E}^{\mathbb{Q}}[A_{n+1} \mid \mathcal{F}_n] \\ &= H_n. \end{aligned}$$

Choosing the already-constructed optimal rule at each child attains H_n. Immediate exercise attains G_n. Hence the best available value is their maximum. The decision to exercise where $G_n \geq H_n$ and otherwise follow the child rule is exactly first entry into $A = G$. This proves both the identity and attainment. □

2.7.3 The Snell envelope and its martingale interpretation

Definition 2.12 (Discounted exercise payoff and value)**.** Set $Y_n = R^{-n}G_n$ and $Z_n = R^{-n}A_n$. Then

$$Z_N = Y_N, \qquad Z_n = \max\{Y_n, \mathbb{E}^{\mathbb{Q}}[Z_{n+1} \mid \mathcal{F}_n]\}.$$

The process Z is called the Snell envelope of Y.

Proposition 2.4 (Smallest dominating supermartingale)**.** *The discounted American value Z is a supermartingale and dominates Y. It is the smallest supermartingale with this property. Moreover, $Z_{k\wedge\tau_0^*}$ is a martingale.*

Proof. The recursion directly gives $Z_n \geq Y_n$ and $Z_n \geq \mathbb{E}^{\mathbb{Q}}[Z_{n+1} \mid \mathcal{F}_n]$. If M is another supermartingale with $M_n \geq Y_n$, backward induction gives

$$\begin{aligned} M_N &\geq Y_N = Z_N, \\ M_n &\geq \max\{Y_n, \mathbb{E}^{\mathbb{Q}}[M_{n+1} \mid \mathcal{F}_n]\} \\ &\geq \max\{Y_n, \mathbb{E}^{\mathbb{Q}}[Z_{n+1} \mid \mathcal{F}_n]\} = Z_n. \end{aligned}$$

Before τ_0^*, the strict inequality $A_n > G_n$ implies $A_n = H_n$, and hence $Z_n = \mathbb{E}^{\mathbb{Q}}[Z_{n+1} \mid \mathcal{F}_n]$. After stopping, the process is constant. Therefore

$$\begin{aligned} &\mathbb{E}^{\mathbb{Q}}[Z_{(n+1)\wedge\tau_0^*} - Z_{n\wedge\tau_0^*} \mid \mathcal{F}_n] \\ &\quad = \mathbb{I}_{\{\tau_0^* > n\}}\mathbb{E}^{\mathbb{Q}}[Z_{n+1} - Z_n \mid \mathcal{F}_n] = 0. \end{aligned}$$

This proves the stopped martingale identity directly, without invoking a general optional-stopping theorem. □

Remark 2.11 (Why the unstopped American value is not generally a martingale). At a node where exercising is preferable, $A_n > H_n$. The discounted value then has a strictly negative conditional drift if one artificially continues the unexercised contract. Equality of discounted expectations holds along optimal continuation and for the process stopped upon optimal exercise.

2.7.4 Why the recursion gives the arbitrage-free price

Proposition 2.5 (Buyer value and writer's covering cost)**.** *The American contract has price A_n in the complete finite tree.*

Proof. A writer starting with A_n can pay G_n if exercise occurs immediately. If not, the writer purchases the one-step hedge of the two child values, costing $H_n \leq A_n$, and puts the surplus $A_n - H_n \geq 0$ in the bank. At the next date the hedge provides exactly the required child value, with the surplus as an additional nonnegative reserve. Repeating this construction covers any holder exercise rule. If the contract were priced above A_n, selling it and buying this covering strategy would create an initial receipt and no future loss.

For the other direction, the optimal stopping rule attains the value A_n. Its prescribed dated payoff is replicable by the construction preceding the theorem. If the American contract cost less, buy it, commit to that rule, and sell its cash-flow replica. The cash flows cancel at exercise, leaving the initial receipt. Thus a price below A_n also permits arbitrage. □

This proof distinguishes a writer's superhedge against *every* exercise rule from exact replication of the cash flow under a particular optimal rule. The additional reserve keeps the writer's full strategy self-financing; no unrecorded outside funding is used.

2.7.5 The lecture's American put, including exercise decisions

Example 2.5. Use the two-period put data from the preceding section. At maturity the payoffs remain $0, 23.5, 86.5$. At the first up node,

$$\begin{aligned} G_{1,U} &= (160 - 195)^+ = 0, \\ H_{1,U} &= 10.99194959, \\ A_{1,U} &= \max\{0, 10.99194959\} = 10.99194959. \end{aligned}$$

Continue at that node. At the first down node,

$$\begin{aligned} G_{1,D} &= 160 - 105 = 55, \\ H_{1,D} &= 52.61791034, \\ A_{1,D} &= 55. \end{aligned}$$

Exercise at that node. At the root,

$$\begin{aligned} G_0 &= 160 - 150 = 10, \\ H_0 &= R^{-1}\big(q(10.99194959) + (1-q)55\big) \\ &\approx 31.41273788, \\ A_0 &= \max\{10, H_0\} = 31.41273788. \end{aligned}$$

The optimal rule waits at zero, exercises at h after D, and otherwise waits until $2h$. Its value can also be calculated directly with the correct payment dates:

$$\begin{aligned} A_0 &= (1-q)R^{-1}(55) + q(1-q)R^{-2}(23.5) + q^2R^{-2}(0) \\ &\approx 31.41273788. \end{aligned}$$

The American premium is

$$A_0 - p_0 \approx 31.41273788 - 30.29853322 = 1.11420466.$$

The root hedge of the child values is

$$\begin{aligned} \Delta_1^A &= \frac{10.99194959 - 55}{195 - 105} \approx -0.48897834, \\ b_1^A &= A_0 - 150\Delta_1^A \approx 104.75948856. \end{aligned}$$

After a down move, its wealth is exactly 55, funding exercise. After an up move, rebalance to the European put's upper-node hedge.

Remark 2.12 (No premature pruning). Compute continuation values before deleting exercise branches from a diagram. The hypothetical continuation payoff is needed to decide whether exercising is optimal. After that comparison, paths beyond an exercised contract can be omitted from the realized cash-flow picture.

2.8 Dividends: gains processes and event timing

Source coverage: lecture slides 84–98..

2.8.1 Ex-dividend prices versus total returns

A shareholder receives both price changes and dividends; an ordinary option holder does not receive the stock's distributions. Consequently, it is the appropriately discounted *total-return investment*, rather than the raw ex-dividend stock price, that must be a pricing-measure martingale.

Definition 2.13 (Cum-dividend and ex-dividend prices)**.** At a dividend date, let S^- be the price immediately before the payment and S^+ the price immediately after. If the distribution is D, the idealized price adjustment is

$$S^+ = S^- - D.$$

The total value received by the shareholder at the event is $S^+ + D = S^-$.

This is the isolated theoretical adjustment with no simultaneous market news or other price movement. It is not a claim that every observed price change at an actual dividend event equals the dividend exactly.

2.8.2 Continuous proportional yield in a grid model

Theorem 2.11 (Continuous-yield binomial probabilities and hedge)**.** *Assume the ex-dividend price moves from s to us or ds over a period, and dividend reinvestment multiplies the share count by $e^{\delta h}$. Then no-arbitrage is equivalent to*

$$d < e^{(r-\delta)h} < u.$$

The up probability is

$$q_\delta = \frac{e^{(r-\delta)h} - d}{u - d}.$$

A claim with next-node values X_U, X_D has value

$$v = R^{-1}\big(q_\delta X_U + (1 - q_\delta)X_D\big).$$

If a denotes the number of shares purchased at the start and reinvested during the period, its replicating holdings are

$$a = e^{-\delta h}\frac{X_U - X_D}{s(u - d)}, \qquad b = \frac{uX_D - dX_U}{R(u - d)}.$$

Proof. One initial share with reinvested dividends becomes $e^{\delta h}$ shares at the next date. Thus its total-return gross factors are $e^{\delta h}u$ and $e^{\delta h}d$. Apply the no-arbitrage condition to these two tradable returns:

$$\begin{aligned} e^{\delta h}d < R < e^{\delta h}u &\iff d < Re^{-\delta h} < u \\ &\iff d < e^{(r-\delta)h} < u. \end{aligned}$$

The martingale probability must satisfy

$$\begin{aligned} e^{\delta h}(q_\delta u + (1-q_\delta)d) &= R, \\ q_\delta u + (1-q_\delta)d &= e^{(r-\delta)h}, \\ q_\delta &= \frac{e^{(r-\delta)h} - d}{u-d}. \end{aligned}$$

For replication, require

$$\begin{aligned} ae^{\delta h}us + bR &= X_U, \\ ae^{\delta h}ds + bR &= X_D. \end{aligned}$$

Subtracting gives the formula for a; substitution gives the stated b. Its cost simplifies as

$$\begin{aligned} as + b &= e^{-\delta h}\frac{X_U - X_D}{u-d} + R^{-1}\frac{uX_D - dX_U}{u-d} \\ &= R^{-1}\left[\frac{Re^{-\delta h} - d}{u-d}X_U + \frac{u - Re^{-\delta h}}{u-d}X_D\right] \\ &= R^{-1}(q_\delta X_U + (1-q_\delta)X_D). \end{aligned}$$

□

Remark 2.13 (Two different rates in the calculation). The dividend yield changes the probabilities through $r - \delta$. The derivative's dollar payoff is still discounted at r, not at $r - \delta$. Also, the u, d in this formula are factors for the ex-dividend price. Applying a further dividend drop to an already adjusted price tree would count the dividend twice.

Proposition 2.6 (The reinvested stock is the discounted martingale)**.** *Define $\widehat{S}_n = e^{\delta nh}S_n$, the value of the reinvested investment that began with one share. Then*

$$\mathbb{E}^{\mathbb{Q}}[S_{n+1} \mid \mathcal{F}_n] = e^{(r-\delta)h}S_n, \qquad \widehat{S}_n/R^n \text{ is a martingale.}$$

Proof. The first identity follows from the equation for q_δ. For the second,

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[\widehat{S}_{n+1}/R^{n+1} \mid \mathcal{F}_n] &= e^{\delta(n+1)h}R^{-(n+1)}e^{(r-\delta)h}S_n \\ &= e^{\delta nh}R^{-n}S_n = \widehat{S}_n/R^n. \end{aligned}$$

□

Applying the local replication step repeatedly gives the same European and American backward recursions as before, with q_δ replacing q. This grid model permits automatic dividend reinvestment within a step; it does not specify discretionary intraperiod trading or exercise.

Example 2.6 (The lecture's 4% dividend-yield put)**.** Keep $S_0 = 150$, $K = 160$, $N = 2$, $h = 0.25$, $u = 1.3$, $d = 0.7$, and $r = 0.06$, but set $\delta = 0.04$. Then

$$\begin{aligned} q_\delta &= \frac{e^{(0.06-0.04)0.25} - 0.7}{0.6} \approx 0.50835420, \\ v_{1,U} &= e^{-0.015}(1-q_\delta)23.5 \approx 11.38166444, \\ v_{1,D} &= e^{-0.015}(q_\delta 23.5 + (1-q_\delta)86.5) \approx 53.66267779, \\ p_0 &= e^{-0.015}(q_\delta v_{1,U} + (1-q_\delta)v_{1,D}) \approx 31.69001379. \end{aligned}$$

The ex-dividend stock levels and terminal put payoffs are the same specified numbers as before; the pricing probabilities change. This verifies the exercise on slide 91.

2.8.3 Discrete dividends and the order of events

For the next examples, a period's multiplicative move first determines a cum-dividend price. Any scheduled dividend is then paid, leaving the ex-dividend value from which the next period's move begins. Shareholders receive the distribution, and ordinary European options continue on the adjusted stock without receiving it.

Suppose the current ex-dividend stock is s, the next cum-dividend values are us, ds, and the corresponding dividend payments are D_U, D_D. Then

$$S_{n+1}^U + D_U = us,$$
$$S_{n+1}^D + D_D = ds.$$

The shareholder's total one-step payoff therefore still has factors u, d, so

$$q = \frac{R-d}{u-d}.$$

This is why the lecture's discrete-dividend examples keep the same q while changing the price tree. For a claim whose post-event child values are v_U, v_D, hold

$$\Delta = \frac{v_U - v_D}{s(u-d)}, \qquad b = \frac{uv_D - dv_U}{R(u-d)}.$$

The stock plus its dividend and the bank payoff reproduce the child values. A European derivative has no cash distribution at the dividend instant, so its value immediately before and immediately after that isolated known event agrees when the stock argument is adjusted consistently.

Remark 2.14 (American exercise at a dividend event). If exercise is permitted immediately before a dividend, that is a separate decision from exercise immediately afterward. Compare the pre-dividend intrinsic value with the value of retaining the option through the event. A tree that checks only post-dividend exercise omits a potentially valuable opportunity, especially for calls. Always specify whether the dividend precedes or follows the exercise decision at a grid date.

2.8.4 A discrete proportional dividend: all numbers worked out

Example 2.7. Let $S_0 = K = 100$, $N = 3$, $h = 1$, $r = 0$, $u = 1.25$, and $d = 0.8$. Immediately after the move at year two, pay 10% of the cum-dividend price. Then $R = 1$ and

$$q = \frac{1-0.8}{1.25-0.8} = \frac{4}{9}, \qquad 1-q = \frac{5}{9}.$$

At year one, the prices are 125 and 80. At year two,

History	Cum-dividend price	Dividend	Ex-dividend price
UU	156.25	15.625	140.625
UD, DU	100	10	90
DD	64	6.4	57.6

The final stock levels are $175.78125, 112.5, 72, 46.08$, producing call payoffs $75.78125, 12.5, 0, 0$. Working backward,

$$\begin{aligned} v_{2,UU} &= \tfrac{4}{9}(75.78125) + \tfrac{5}{9}(12.5) = 40.625, \\ v_{2,UD} = v_{2,DU} &= \tfrac{4}{9}(12.5) + \tfrac{5}{9}(0) = 5.55555556, \\ v_{2,DD} &= 0, \\ v_{1,U} &= \tfrac{4}{9}(40.625) + \tfrac{5}{9}(5.55555556) = 21.14197531, \\ v_{1,D} &= \tfrac{4}{9}(5.55555556) + \tfrac{5}{9}(0) = 2.46913580, \\ c_0 &= \tfrac{4}{9}(21.14197531) + \tfrac{5}{9}(2.46913580) = 10.76817558. \end{aligned}$$

Because multiplication by 0.9 commutes with the up and down factors, terminal stock values remain $0.9S_0u^jd^{3-j}$. Thus this scheduled proportional-dividend tree still recombines.

2.8.5 A fixed cash dividend: loss of usual recombination

Example 2.8. Use the same model but replace the 10% distribution by ten dollars at year two. The ex-dividend prices at that date are now $146.25, 90, 54$. The six distinct year-three prices arising from these nodes are

Year-two history	Ex price	Next up price	Next down price
UU	146.25	182.8125	117
UD, DU	90	112.5	72
DD	54	67.5	43.2

The corresponding next-step call payoffs are $(82.8125, 17)$, $(12.5, 0)$, and $(0, 0)$. Therefore

$$\begin{aligned} v_{2,UU} &= \tfrac{4}{9}(82.8125) + \tfrac{5}{9}(17) = 46.25, \\ v_{2,UD} = v_{2,DU} &= \tfrac{4}{9}(12.5) = 5.55555556, \\ v_{2,DD} &= 0, \\ v_{1,U} &= \tfrac{4}{9}(46.25) + \tfrac{5}{9}(5.55555556) = 23.64197531, \\ v_{1,D} &= 2.46913580, \\ c_0 &= \tfrac{4}{9}(23.64197531) + \tfrac{5}{9}(2.46913580) = 11.87928669. \end{aligned}$$

The ordinary recombination fails because subtracting a cash amount does not commute with subsequent multiplication. For example,

$$\begin{aligned} S_3(UUD) &= (100u^2 - 10)d = 117, \\ S_3(UDU) &= (100ud - 10)u = 112.5, \\ S_3(UUD) - S_3(UDU) &= 10(u - d) = 4.5. \end{aligned}$$

Both histories have two up moves, yet their final prices differ.

Remark 2.15 (Nonrecombination is not path dependence of the payoff). The call still pays $(S_N - K)^+$ and therefore depends only on the terminal stock price. What fails is the representation of that terminal price by the up count alone. Conversely, even a recombining stock tree can support a genuinely path-dependent option whose value cannot be represented using stock price alone.

Remark 2.16 (Large cash dividends). A fixed dividend larger than a node's cum-dividend stock price would produce a negative ex-dividend value under naive subtraction. That is inconsistent with the nonnegative-stock convention. One must modify the dividend model or the state space rather than silently use the invalid node. The lecture's numerical example has positive ex-dividend prices at every node.

2.9 Path-dependent claims and sufficient state variables

Source coverage: lecture slides 99–110..

2.9.1 Full histories versus augmented states

Definition 2.14 (Path-dependent claim)**.** A terminal claim is path-dependent if it cannot be represented as a function of S_N alone. It may instead be written as $X = \Phi(S_0, \ldots, S_N)$.

The replication and conditional valuation theorems already cover such claims. The difficulty is computational state representation, not a new pricing principle.

Proposition 2.7 (Recursion with an augmented state)**.** *Suppose a state variable Y_n records the information needed for future transitions and the payoff, with $Y_{n+1} = F_n(Y_n, Z_{n+1})$ and $X = g(Y_N)$. Then*

$$\begin{aligned} v_N(y) &= g(y), \\ v_n(y) &= R^{-1}\big[q v_{n+1}(F_n(y,u)) + (1-q) v_{n+1}(F_n(y,d))\big]. \end{aligned}$$

Histories can be merged whenever they have the same time and sufficient state y.

Proof. Conditional on $Y_n = y$, the next state has the two displayed values with probabilities $q, 1-q$, and the future transition rules and payoff no longer need the discarded history. Apply one-step conditional valuation, starting from $g(Y_N)$ and inducting backward. □

For an Asian claim, use current stock and running sum; for a barrier, use current stock and hit status; for a lookback, use current stock and running extrema. A full 2^N-path expansion is always a valid finite-tree method, but it is not always computationally necessary. This refines the lecture's instruction to draw every path.

2.9.2 Common three-period model and all eight histories

The three exotic examples use

$$S_0 = 200, \quad N = 3, \quad h = 1, \quad r = 0.05, \quad u = 1.2, \quad d = 0.9,$$

with

$$q = \frac{e^{0.05} - 0.9}{0.3} \approx 0.50423699, \qquad D_1 = e^{-0.05}.$$

The path data are

Path	S_1	S_2	S_3	Average $\bar{S}$	$\min_{0\le n\le 3} S_n$	$\max_{0\le n\le 3} S_n$
UUU	240	288	345.6	291.2	200	345.6
UUD	240	288	259.2	262.4	200	288
UDU	240	216	259.2	238.4	200	259.2
UDD	240	216	194.4	216.8	194.4	240
DUU	180	216	259.2	218.4	180	259.2
DUD	180	216	194.4	196.8	180	216
DDU	180	162	194.4	178.8	162	200
DDD	180	162	145.8	162.6	145.8	200

Here $\bar{S} = (S_1 + S_2 + S_3)/3$ excludes the initial stock price, exactly as in the lecture's Asian contract. The extrema include time zero and maturity. A contract using different observation dates is a different claim.

2.9.3 Asian options and the running-average state

Definition 2.15 (Arithmetic-average Asian option)**.** For specified observation dates $1, \dots, N$, define $\bar{S} = N^{-1} \sum_{i=1}^{N} S_i$. An Asian call pays $(\bar{S} - K)^+$; an Asian put pays $(K - \bar{S})^+$.

Set $L_n = \sum_{i=1}^{n} S_i$ with $L_0 = 0$. A sufficient state is (S_n, L_n), since

$$(s, \ell) \longmapsto (us, \ell + us) \quad \text{on up},$$
$$(s, \ell) \longmapsto (ds, \ell + ds) \quad \text{on down}.$$

For a put, $v_N(s, \ell) = (K - \ell/N)^+$.

Example 2.9 (The lecture's Asian put)**.** Take $K = 200$ in the common model. Only paths DUD, DDU, DDD have positive payoffs:

$$X(DUD) = 200 - 196.8 = 3.2,$$
$$X(DDU) = 200 - 178.8 = 21.2,$$
$$X(DDD) = 200 - 162.6 = 37.4.$$

All four paths beginning with U pay zero, and DUU also pays zero. Backward induction gives

$$\begin{aligned} v_{2,DU} &= D_1(q(0) + (1-q)3.2) \approx 1.50906997, \\ v_{2,DD} &= D_1(q21.2 + (1-q)37.4) \approx 27.80573051, \\ v_{1,D} &= D_1(qv_{2,DU} + (1-q)v_{2,DD}) \approx 13.83656571, \\ v_{1,U} &= 0, \\ v_0 &= D_1(1-q)v_{1,D} \approx 6.52510805. \end{aligned}$$

The direct path calculation verifies the result:

$$\begin{aligned} v_0 &= e^{-0.15}\left[q(1-q)^2(3.2) + q(1-q)^2(21.2) + (1-q)^3(37.4)\right] \\ &= e^{-0.15}\left[q(1-q)^2(24.4) + (1-q)^3(37.4)\right] \\ &\approx 6.52510805. \end{aligned}$$

Notice that UDD, DUD, DDU all end at 194.4 but have different averages and payoffs. Multiplying one terminal payoff by a binomial coefficient would be incorrect.

2.9.4 Barrier monitoring and knock-in/knock-out parity

Definition 2.16 (Discretely monitored down-and-out call)**.** With lower barrier L and no rebate, define the first monitored hit

$$\tau_L = \inf\{n \in \{0, \dots, N\} : S_n \le L\},$$

with $\inf \varnothing = \infty$. The payoff is

$$X_{\text{out}} = (S_N - K)^+ \mathbb{I}_{\{\tau_L > N\}}.$$

The matching no-rebate down-and-in call pays

$$X_{\text{in}} = (S_N - K)^+ \mathbb{I}_{\{\tau_L \le N\}}.$$

The hit flag $I_n = \mathbb{I}_{\{\tau_L \le n\}}$ is sufficient additional information. Once hit, a no-rebate knock-out value is zero. For an upper barrier, replace the event $S_n \le L$ by $S_n \ge H$. Whether equality counts as a hit, the observation dates, and any rebate amount and payment date must be specified.

Proposition 2.8 (In-out parity)**.** *For matching monitoring conventions and no rebates,*

$$X_{\text{in}} + X_{\text{out}} = (S_N - K)^+, \qquad v_{\text{in}} + v_{\text{out}} = c_0.$$

Proof. The hit and no-hit events partition the sample space, so their indicators sum to one. Add the payoffs and apply linear pricing. □

Example 2.10 (The lecture's barrier call)**.** Take $K = 200$ and lower barrier $L = 190$. Any path beginning with D hits immediately because $S_1 = 180$. The surviving positive terminal payoffs are

$$X(UUU) = 145.6, \qquad X(UUD) = 59.2, \qquad X(UDU) = 59.2.$$

Path UDD survives monitoring but finishes at $194.4 < K$, so its call payoff is zero. Thus

$$\begin{aligned} v_{2,UU} &= D_1\big(q145.6 + (1-q)59.2\big) \approx 97.75411510, \\ v_{2,UD} &= D_1\big(q59.2 + (1-q)0\big) \approx 28.39498754, \\ v_{1,U} &= D_1\big(qv_{2,UU} + (1-q)v_{2,UD}\big) \approx 60.27791055, \\ v_{1,D} &= 0, \\ v_0 &= D_1 q v_{1,U} \approx 28.91200201. \end{aligned}$$

Equivalently,

$$v_0 = e^{-0.15}\left[145.6q^3 + 2(59.2)q^2(1-q)\right].$$

The factor two counts UUD and UDU. The third path ending at 259.2, namely DUU, was knocked out and must not be counted.

Remark 2.17 (No inference about between-date barrier crossings). The model only observes grid-date prices. A continuously monitored barrier could be hit between two dates even if both endpoints are above it, but no intraperiod path is present here. A continuous-monitoring correction requires additional modelling; it is not supplied by the discrete tree itself.

2.9.5 Lookback contracts and running extrema

Definition 2.17 (Lookback payoffs)**.** For observation times including zero and maturity, let

$$m_N = \min_{0\le n\le N} S_n, \qquad M_N = \max_{0\le n\le N} S_n.$$

The four common payoffs are

$$\begin{aligned} \text{floating-strike call: } & S_N - m_N, \\ \text{floating-strike put: } & M_N - S_N, \\ \text{fixed-strike call: } & (M_N - K)^+, \\ \text{fixed-strike put: } & (K - m_N)^+. \end{aligned}$$

The floating payoffs are automatically nonnegative because maturity is included among the observation times. For a floating-strike put, (S_n, M_n) is sufficient, with next maximum $\max\{M_n, S_{n+1}\}$.

Example 2.11 (The lecture's floating-strike lookback put)**.** Using the common path table, the terminal payoffs in lexicographic path order are

Path	UUU	UUD	UDU	UDD	DUU	DUD	DDU	DDD
$M_3 - S_3$	0	28.8	0	45.6	0	21.6	5.6	54.2

For example, on UDD the running maximum is 240, so the payoff is $240 - 194.4 = 45.6$. On DDD, the maximum is the initial price 200, giving $200 - 145.8 = 54.2$. Backward induction gives

$$\begin{aligned} v_{2,UU} &= D_1(1-q)28.8 \approx 13.58162970, \\ v_{2,UD} &= D_1(1-q)45.6 \approx 21.50424703, \\ v_{2,DU} &= D_1(1-q)21.6 \approx 10.18622228, \\ v_{2,DD} &= D_1\big(q5.6 + (1-q)54.2\big) \approx 28.24588490. \end{aligned}$$

The first-period values and root price are

$$\begin{aligned} v_{1,U} &= D_1\big(qv_{2,UU} + (1-q)v_{2,UD}\big) \approx 16.65542826, \\ v_{1,D} &= D_1\big(qv_{2,DU} + (1-q)v_{2,DD}\big) \approx 18.20608888, \\ v_0 &= D_1\big(qv_{1,U} + (1-q)v_{1,D}\big) \approx 16.57440074. \end{aligned}$$

As a second check,

$$\begin{aligned} v_0 = e^{-0.15}\big[&28.8q^2(1-q) + (45.6 + 21.6 + 5.6)q(1-q)^2 \\ &+ 54.2(1-q)^3\big] \approx 16.57440074. \end{aligned}$$

In particular, $v_{2,UD} \neq v_{2,DU}$ even though both stock prices are 216: their running maxima are 240 and 216 respectively.

2.10 General one-period finite-state markets

Source coverage: lecture slides 111–125..

2.10.1 The payoff matrix and portfolio cash flows

Definition 2.18 (Finite-state market)**.** Let $\Omega = \{\omega_1, \ldots, \omega_M\}$ with $\mathbb{P}(\{\omega_j\}) > 0$. There are L traded securities, with initial price row vector

$$s = (s_1, \ldots, s_L) \in \mathbb{R}^{1\times L}$$

and terminal payoff matrix $\mathsf{A} \in \mathbb{R}^{M\times L}$, where A_{ji} is asset i's payoff in state j. All contractual payments are included in that terminal payoff; intermediate payments, if any, must first be accounted for consistently.

There need not be a risk-free asset. The use of L for the number of securities avoids confusing it with the N time steps used earlier. A column is one security's payoff vector; a row is the list of all security payoffs in one future state.

Definition 2.19 (Portfolio in matrix form)**.** A portfolio is a column vector $\theta \in \mathbb{R}^L$. Its initial cost is $s\theta$ and its terminal payoff vector is $\mathsf{A}\theta$. Explicitly,

$$\mathsf{A}\theta = \begin{pmatrix} \sum_{i=1}^{L} \mathsf{A}_{1i}\theta_i \\ \sum_{i=1}^{L} \mathsf{A}_{2i}\theta_i \\ \vdots \\ \sum_{i=1}^{L} \mathsf{A}_{Mi}\theta_i \end{pmatrix}.$$

Example 2.12 (The lecture's three-asset portfolio)**.** Let

$$s = \begin{pmatrix} 1 & 20 & 50 \end{pmatrix}, \qquad \mathsf{A} = \begin{pmatrix} 1.1 & 25 & 100 \\ 1.1 & 15 & 20 \\ 1.1 & 30 & 30 \end{pmatrix}, \qquad \theta = \begin{pmatrix} 200 \\ -10 \\ 3 \end{pmatrix}.$$

The initial cost is

$$\begin{aligned} V_0 &= 1(200) + 20(-10) + 50(3) \\ &= 200 - 200 + 150 = 150. \end{aligned}$$

The terminal payoff vector is

$$\begin{aligned} \mathsf{A}\theta &= \begin{pmatrix} 1.1(200) + 25(-10) + 100(3) \\ 1.1(200) + 15(-10) + 20(3) \\ 1.1(200) + 30(-10) + 30(3) \end{pmatrix} \\ &= \begin{pmatrix} 270 \\ 130 \\ 10 \end{pmatrix}. \end{aligned}$$

These are values in three alternative future states, not three payments received together. The portfolio is not an arbitrage merely because every terminal value is positive: it costs 150 initially.

2.10.2 Arbitrage when no bank account is assumed

Definition 2.20 (Arbitrage as a nonnegative cash-flow vector)**.** In the general market, an arbitrage is a portfolio with

$$-s\theta \geq 0, \qquad \mathsf{A}\theta \geq 0,$$

and at least one of these $M+1$ cash-flow components strictly positive. Here $\mathsf{A}\theta \geq 0$ means componentwise nonnegativity.

This includes a strictly positive initial receipt with identically zero terminal obligations. When a positive bank account is available, that receipt can be invested to produce the more familiar strictly positive terminal gain. Without such an asset, the initial-receipt case must be included explicitly to state the finite-state fundamental theorem correctly.

Remark 2.18 (A necessary refinement of slide 119). The lecture's definition requires at least one strictly positive terminal component. Without a bank account, that convention misses, for example, shorting an asset priced at one whose terminal payoff is identically zero. Our definition counts the initial receipt as a genuine gain. This is the convention used in the full state-price theorem below.

2.10.3 State-price vectors

Definition 2.21 (Strictly positive state prices)**.** A state-price vector is a row vector $\psi = (\psi_1, \ldots, \psi_M)$ with $\psi_j > 0$ for every j and

$$\boxed{s = \psi\mathsf{A}.}$$

Equivalently, for each traded asset i,

$$s_i = \sum_{j=1}^{M} \psi_j \mathsf{A}_{ji}.$$

A state-price coefficient prices one unit delivered in a particular state, when such a state claim is attainable. The coefficients need not sum to one: they are dollar prices, not yet probabilities.

Proposition 2.9 (Positive state prices rule out arbitrage)**.** *If a strictly positive state-price vector exists, the market is arbitrage-free.*

Proof. For every portfolio,

$$s\theta = \psi\mathsf{A}\theta = \sum_{j} \psi_j (\mathsf{A}\theta)_j.$$

If terminal payoffs are nonnegative and at least one is positive, this cost is positive. If every terminal payoff is zero, the cost is zero, so no strictly positive initial receipt is possible either. Both forms of arbitrage are excluded. □

2.10.4 A full finite-dimensional proof of the converse

The next lemma proves the needed separation fact directly using Euclidean projection. This supplies the converse omitted from the lecture without requiring a general infinite-dimensional separation theorem.

Lemma 2.3 (Strictly positive vector orthogonal to a subspace)**.** *Let $\mathcal{L}$ be a linear subspace of $\mathbb{R}^m$. Then*

$$\mathcal{L} \cap \mathbb{R}^m_+ = \{0\}$$

if and only if there exists $y \in \mathcal{L}^\perp$ with every coordinate strictly positive.

Proof. If such a y exists, any nonzero $\ell \in \mathbb{R}^m_+$ has $y \cdot \ell > 0$ and therefore cannot belong to $\mathcal{L}$.

Conversely, assume the intersection is trivial. Define the simplex

$$K = \{x \in \mathbb{R}^m : x_i \geq 0,\ \sum_{i=1}^m x_i = 1\}.$$

It is compact, convex, and disjoint from $\mathcal{L}$. Let P be orthogonal projection onto $\mathcal{L}^\perp$. The continuous function $\|Px\|^2$ attains its minimum at some $x^* \in K$. Set $y = Px^*$. This vector is nonzero: otherwise $x^* \in \mathcal{L} \cap K$, contradicting disjointness.

For any $x \in K$ and $\varepsilon \in [0,1]$, the point $x^* + \varepsilon(x - x^*)$ lies in K. Minimality implies

$$\begin{aligned} 0 &\leq \|P[x^* + \varepsilon(x - x^*)]\|^2 - \|Px^*\|^2 \\ &= 2\varepsilon\, y \cdot P(x - x^*) + \varepsilon^2 \|P(x - x^*)\|^2. \end{aligned}$$

Divide by $\varepsilon > 0$ and let $\varepsilon \downarrow 0$. Because $y \in \mathcal{L}^\perp$,

$$y \cdot (x - x^*) = y \cdot P(x - x^*) \geq 0.$$

Also $x^* - Px^* \in \mathcal{L}$, so

$$y \cdot x^* = y \cdot Px^* = \|y\|^2 > 0.$$

Choose $x = e_i$, the ith coordinate vector, which belongs to K. Then

$$y_i = y \cdot e_i \geq y \cdot x^* = \|y\|^2 > 0.$$

Thus every coordinate of y is strictly positive. □

Theorem 2.12 (First fundamental theorem for a finite one-period market)**.** *Under the cash-flow definition of arbitrage above, the market is arbitrage-free if and only if a strictly positive state-price vector exists.*

Proof. Sufficiency was proved above. For necessity, consider the subspace of all obtainable cash-flow vectors:

$$\mathcal{L} = \left\{ \begin{pmatrix} -s\theta \\ \mathsf{A}\theta \end{pmatrix} : \theta \in \mathbb{R}^L \right\} \subseteq \mathbb{R}^{M+1}.$$

No-arbitrage says precisely that $\mathcal{L} \cap \mathbb{R}^{M+1}_+ = \{0\}$. The lemma gives a strictly positive vector $y = (y_0, y_1, \ldots, y_M)$ orthogonal to $\mathcal{L}$. Hence for every θ,

$$\begin{aligned} 0 &= -y_0 s\theta + (y_1, \ldots, y_M)\mathsf{A}\theta, \\ y_0 s &= (y_1, \ldots, y_M)\mathsf{A}. \end{aligned}$$

Because $y_0 > 0$, define $\psi_j = y_j / y_0 > 0$. Then $s = \psi\mathsf{A}$, as required. □

Remark 2.19 (This proof also explains strict positivity). The separating vector is positive in every time/state cash-flow coordinate. Nonnegative weights would not exclude profit in a state assigned weight zero. The geometric distinction matches the boundary-arbitrage issue already seen in the binomial model.

2.10.5 Solving the lecture's three-state state-price family

Example 2.13. Let

$$s = \begin{pmatrix} 1 & 1 \end{pmatrix}, \qquad \mathsf{A} = \begin{pmatrix} 2 & 0 \\ 2 & 4 \\ 2 & 3 \end{pmatrix}.$$

Write $\psi = (a, b, x)$. The state-price equations are

$$\begin{aligned} 2a + 2b + 2x &= 1, \\ 4b + 3x &= 1. \end{aligned}$$

Thus

$$\begin{aligned} b &= \frac{1}{4} - \frac{3}{4}x, \\ a &= \frac{1}{2} - b - x \\ &= \frac{1}{2} - \left(\frac{1}{4} - \frac{3}{4}x\right) - x \\ &= \frac{1}{4} - \frac{1}{4}x. \end{aligned}$$

Strict positivity requires

$$\begin{aligned} x &> 0, \\ \frac{1}{4} - \frac{3}{4}x > 0 &\iff x < \frac{1}{3}, \\ \frac{1}{4} - \frac{1}{4}x > 0 &\iff x < 1. \end{aligned}$$

Therefore all state-price vectors are

$$\boxed{\psi(x) = \left(\frac{1}{4} - \frac{1}{4}x,\ \frac{1}{4} - \frac{3}{4}x,\ x\right), \qquad 0 < x < \frac{1}{3}.}$$

Their existence proves no-arbitrage, while their nonuniqueness will signal incompleteness.

Example 2.14 (Recovering binomial state prices)**.** For normalized stock price $S_0 = 1$,

$$s = (1, 1), \qquad \mathsf{A} = \begin{pmatrix} R & u \\ R & d \end{pmatrix}.$$

If $\psi = (a, b)$, then

$$\begin{aligned} R(a + b) &= 1, \\ ua + db &= 1. \end{aligned}$$

Substitute $b = R^{-1} - a$:

$$\begin{aligned}(u-d)a &= 1 - dR^{-1},\\ a &= R^{-1}\frac{R-d}{u-d} = R^{-1}q,\\ b &= R^{-1}(1-q).\end{aligned}$$

Both coefficients are positive exactly when $d < R < u$.

2.11 Pricing new claims and characterizing completeness

Source coverage: lecture slides 125–132..

2.11.1 Adding a claim as a new traded asset

Theorem 2.13 (All arbitrage-free prices of a claim)**.** *Suppose the original finite-state market is arbitrage-free, and let $X \in \mathbb{R}^M$ be a new claim. A proposed price x_0 keeps the enlarged market arbitrage-free if and only if*

$$x_0 = \psi X$$

for at least one strictly positive state-price vector ψ of the original market. Hence the price set is

$$\mathcal{I}(X) = \{\psi X : \psi > 0,\ \psi \mathsf{A} = s\}.$$

Proof. The enlarged price vector is (s, x_0) and the enlarged payoff matrix is $[\mathsf{A} \mid X]$. By the finite-state fundamental theorem, no-arbitrage is equivalent to existence of $\psi > 0$ such that

$$\begin{aligned}(s, x_0) &= \psi[\mathsf{A} \mid X]\\ &= (\psi\mathsf{A}, \psi X).\end{aligned}$$

The first coordinates state that ψ prices the original assets; the last coordinate gives $x_0 = \psi X$. □

Remark 2.20 (The quantifier is "there exists"). A new claim can select one pricing functional from the original market's family. It need not have the same price under every original state-price vector unless it is already attainable. When several new claims are added together, one common positive state-price vector must price all of them simultaneously.

2.11.2 An open interval of call prices and its excluded endpoints

Example 2.15 (The lecture's incomplete-market call)**.** In the preceding three-state market, a call on the second asset with strike 2 pays

$$X = \begin{pmatrix} (0-2)^+ \\ (4-2)^+ \\ (3-2)^+ \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}.$$

For the state-price family,

$$\begin{aligned} x_0(x) &= \psi(x)X \\ &= 2\left(\frac{1}{4} - \frac{3}{4}x\right) + x \\ &= \frac{1}{2} - \frac{1}{2}x. \end{aligned}$$

As x runs through $(0, 1/3)$, the arbitrage-free prices run through

$$\boxed{\mathcal{I}(X) = \left(\frac{1}{3}, \frac{1}{2}\right).}$$

The interval is open because strict state-price positivity is required.

The endpoints can also be excluded by explicit trades. One-third of the second asset pays $(0, 4/3, 1)$ and costs $1/3$. Therefore

$$X - \tfrac{1}{3}\mathsf{A}_{\cdot,2} = \begin{pmatrix} 0 \\ 2/3 \\ 0 \end{pmatrix}.$$

At a call price of $1/3$, buying the call and shorting one-third of the asset produces this positive-in-one-state payoff at zero cost. Any still lower price also gives an initial receipt.

Half of the second asset pays $(0, 2, 3/2)$ and costs $1/2$. Thus

$$\tfrac{1}{2}\mathsf{A}_{\cdot,2} - X = \begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix}.$$

At call price $1/2$, buying half the asset and selling the call is arbitrage. Any higher price also produces an initial receipt.

Example 2.16 (Adding the call at price $2/5$)**.** Impose $x_0 = 2/5$. Then

$$\begin{aligned} \frac{2}{5} &= \frac{1}{2} - \frac{1}{2}x, \\ x &= \frac{1}{5}, \\ \psi &= \left(\frac{1}{5}, \frac{1}{10}, \frac{1}{5}\right). \end{aligned}$$

These are all state-price vectors in the enlarged market: there is exactly one. Its payoff matrix is

$$\mathsf{A}_{\text{new}} = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 4 & 2 \\ 2 & 3 & 1 \end{pmatrix}, \qquad \det(\mathsf{A}_{\text{new}}) = 2(4-6) = -4 \neq 0.$$

Adding this independently priced payoff completes the market.

2.11.3 Rank, attainability, and redundant portfolios

Theorem 2.14 (Rank criterion for completeness)**.** *An arbitrage-free market with M states is complete if and only if*

$$\operatorname{rank}(\mathsf{A}) = M.$$

Proof. A claim $X \in \mathbb{R}^M$ is attainable exactly when the linear system $\mathsf{A}\theta = X$ has a solution. Every claim is attainable exactly when the column space of A equals $\mathbb{R}^M$. This is equivalent to rank M. □

Corollary 2.2 (Asset count is necessary, but rank is decisive)**.** *Completeness requires $L \geq M$. Having at least as many assets as states is not sufficient if the asset payoff columns are linearly dependent and fail to span $\mathbb{R}^M$.*

Remark 2.21 (Unique price does not imply a unique replicating portfolio). If $\mathsf{A}\theta = \mathsf{A}\widetilde{\theta} = X$, then $\theta - \widetilde{\theta} \in \ker \mathsf{A}$. For any state-price vector,

$$s(\theta - \widetilde{\theta}) = \psi\mathsf{A}(\theta - \widetilde{\theta}) = 0.$$

So both replicas have the same cost even if their holdings differ. Redundant securities allow nonunique hedges. A square invertible payoff matrix gives unique holdings; full row rank alone gives existence, not uniqueness, when there are more assets than states.

2.11.4 The second fundamental theorem, including the positivity step

Theorem 2.15 (Completeness is equivalent to unique state prices)**.** *An arbitrage-free finite one-period market is complete if and only if its strictly positive state-price vector is unique.*

Proof. No-arbitrage guarantees at least one positive solution ψ to $\psi\mathsf{A} = s$.

If the market is complete, $\operatorname{rank}(\mathsf{A}) = M$, so $\ker(\mathsf{A}^\mathsf{T}) = \{0\}$. If $\widetilde{\psi}$ were another solution, then

$$\mathsf{A}^\mathsf{T}(\widetilde{\psi} - \psi)^\mathsf{T} = 0,$$

forcing $\widetilde{\psi} = \psi$.

Conversely, if $\operatorname{rank}(\mathsf{A}) < M$, choose a nonzero column vector $z \in \ker(\mathsf{A}^\mathsf{T})$. Thus $z^\mathsf{T}\mathsf{A} = 0$. Select

$$0 < \varepsilon < \min_{j:z_j \neq 0} \frac{\psi_j}{|z_j|}.$$

The minimum is strictly positive because there are finitely many coordinates and every $\psi_j > 0$. Then both

$$\psi^+ = \psi + \varepsilon z^\mathsf{T}, \qquad \psi^- = \psi - \varepsilon z^\mathsf{T}$$

are strictly positive and satisfy

$$\psi^\pm\mathsf{A} = \psi\mathsf{A} \pm \varepsilon z^\mathsf{T}\mathsf{A} = s.$$

They are distinct, contradicting uniqueness. Hence uniqueness implies rank M and completeness. □

Remark 2.22 (Why the perturbation argument is needed). Uniqueness among unrestricted linear-system solutions is not automatically the same as uniqueness among positive solutions. Starting with a strictly positive solution and using a sufficiently small kernel perturbation bridges this gap. This is the missing detail in a bare rank-nullity proof.

2.11.5 An individual claim has a unique price exactly when it is attainable

Proposition 2.10 (Claim-level uniqueness)**.** *In an arbitrage-free finite market, $\mathcal{I}(X)$ is a singleton if and only if X belongs to the column space of A.*

Proof. If $X = \mathsf{A}\theta$, every state-price vector gives

$$\psi X = \psi \mathsf{A}\theta = s\theta,$$

so the price is unique.

Conversely, if X is not in the column space, decompose it orthogonally as $X = X_{\parallel} + z$, where $X_{\parallel}$ lies in that column space and $z \neq 0$ is orthogonal to it. Then

$$\mathsf{A}^{\mathsf{T}} z = 0, \qquad z^{\mathsf{T}} X = \|z\|^2 > 0.$$

Starting from any positive ψ, choose a small ε as in the preceding proof. The two positive state-price vectors $\psi \pm \varepsilon z^{\mathsf{T}}$ give prices differing by

$$\begin{aligned}(\psi + \varepsilon z^{\mathsf{T}})X - (\psi - \varepsilon z^{\mathsf{T}})X &= 2\varepsilon z^{\mathsf{T}} X \\ &= 2\varepsilon \|z\|^2 > 0.\end{aligned}$$

Thus the price is not unique. □

2.12 Risk-neutral measures and a change of numeraire

Source coverage: lecture slides 133–136..

2.12.1 Normalizing state prices with a bank account

Theorem 2.16 (State prices and risk-neutral probabilities)**.** *Suppose a traded bank account has initial price one and terminal payoff $R > 0$ in every state. Then positive state-price vectors correspond one-to-one with equivalent risk-neutral measures through*

$$q_j = R\psi_j, \qquad \psi_j = R^{-1} q_j.$$

For every traded payoff X priced by that vector,

$$x_0 = \psi X = R^{-1} \mathbb{E}^{\mathbb{Q}}[X].$$

Proof. The state-price equation for the bank account gives

$$1 = \sum_{j=1}^{M} \psi_j R,$$

$$\sum_{j=1}^{M} q_j = 1.$$

Every $q_j > 0$, so these are probabilities equivalent to the physical measure. For any asset i,

$$\begin{aligned} s_i &= \sum_j \psi_j \mathsf{A}_{ji} \\ &= R^{-1} \sum_j q_j \mathsf{A}_{ji} = R^{-1} \mathbb{E}^{\mathbb{Q}}[S_1^{(i)}]. \end{aligned}$$

Conversely, positive probabilities pricing all assets by this rule define $\psi_j = R^{-1} q_j$, a positive state-price vector. □

If the market is incomplete, there may be several such measures. Each gives the same value for attainable claims; a new nonattainable claim's quote selects those measures consistent with that quote.

2.12.2 A general strictly positive traded numeraire

Definition 2.22 (Numeraire)**.** A numeraire is a traded asset or self-financing portfolio with strictly positive initial value N_0 and strictly positive terminal values $N_1(\omega_j)$. Prices measured in units of this asset are $S^{(i)}/N$.

Strict positivity is indispensable: division by a zero or sign-changing asset does not provide a valid common unit of account.

Theorem 2.17 (Numeraire pricing formula)**.** *Given a positive state-price vector ψ and a strictly positive traded numeraire, define*

$$q_j^N = \frac{\psi_j N_1(\omega_j)}{N_0}.$$

Then $q_j^N > 0$, $\sum_j q_j^N = 1$, and every priced claim satisfies

$$\boxed{x_0 = N_0 \mathbb{E}^{\mathbb{Q}^N}\left[\frac{X}{N_1}\right].}$$

In particular, each traded asset satisfies

$$\frac{s_i}{N_0} = \mathbb{E}^{\mathbb{Q}^N}\left[\frac{S_1^{(i)}}{N_1}\right].$$

Proof. Since the numeraire itself is traded, $N_0 = \sum_j \psi_j N_1(\omega_j)$. Therefore

$$\sum_j q_j^N = \frac{1}{N_0} \sum_j \psi_j N_1(\omega_j) = 1.$$

For a priced payoff,

$$\begin{aligned} N_0 \mathbb{E}^{\mathbb{Q}^N}[X/N_1] &= N_0 \sum_j \frac{\psi_j N_1(\omega_j)}{N_0} \frac{X(\omega_j)}{N_1(\omega_j)} \\ &= \sum_j \psi_j X(\omega_j) = x_0. \end{aligned}$$

Apply this identity to each traded security. □

Remark 2.23 (Normalized prices must stay normalized). The formula $x_0 = \mathbb{E}[X_1]$ is correct only when both values have already been expressed in numeraire units. In currency units, the factors N_0 and $1/N_1$ in the boxed formula cannot be omitted. Choosing a numeraire changes both the unit of account and the associated pricing measure.

Example 2.17 (Bank and stock numeraires in the binomial model)**.** For $S_0 = 1$, the state prices are $\psi_U = R^{-1}q$ and $\psi_D = R^{-1}(1-q)$. The bank numeraire has $N_0 = 1$, $N_1 = R$, so

$$\mathbb{Q}^B(U) = q, \qquad \mathbb{Q}^B(D) = 1 - q.$$

With the non-dividend stock as numeraire, $N_0 = 1$, $N_1(U) = u$, and $N_1(D) = d$. Hence

$$\mathbb{Q}^S(U) = \frac{qu}{R}, \qquad \mathbb{Q}^S(D) = \frac{(1-q)d}{R}.$$

They sum to one because $qu + (1-q)d = R$. A claim's price is unchanged:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^S}[X/S_1] &= \frac{qu}{R}\frac{X_U}{u} + \frac{(1-q)d}{R}\frac{X_D}{d} \\ &= R^{-1}(qX_U + (1-q)X_D).\end{aligned}$$

The probabilities changed, but so did the payoff's units.

2.12.3 What extends to a general finite multiperiod market

With a finite tree, predictable strategies, self-financing trading, and a strictly positive traded numeraire, the same ideas extend node by node. An equivalent measure under which all numeraire-denominated traded prices are martingales rules out arbitrage: every self-financing portfolio's normalized wealth is then a martingale, so a zero-cost nonnegative terminal gain is impossible.

If every local contingent payoff can be spanned, backward induction gives dynamic completeness and a unique equivalent martingale measure. In a complete such market, for a terminal claim X,

$$\pi_n(X) = N_n \mathbb{E}^{\mathbb{Q}^N}[X/N_N \mid \mathcal{F}_n].$$

In an incomplete market, this formula remains valid for attainable claims under every compatible measure; arbitrary new claims can have a range of admissible prices. The finite binomial model is the fully proved special case developed in this chapter. General infinite-state or continuous-time fundamental theorems require additional technical hypotheses and are not being asserted here without them.

2.13 Reference formulas, numerical checks, and source map

2.13.1 Pricing and hedging formulas

For a non-dividend binomial step from current stock s,

$$R = e^{rh}, \qquad d < R < u, \qquad q = \frac{R-d}{u-d},$$
$$v = R^{-1}(qv_U + (1-q)v_D), \qquad \Delta = \frac{v_U - v_D}{s(u-d)},$$
$$b = \frac{uv_D - dv_U}{R(u-d)}, \qquad v = \Delta s + b.$$

For a terminal claim on the full tree,

$$\pi_n(X) = R^{-(N-n)}\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_n].$$

For a grid-exercisable claim,

$$A_N = G_N, \qquad A_n = \max\{G_n, R^{-1}\mathbb{E}^{\mathbb{Q}}[A_{n+1} \mid \mathcal{F}_n]\}.$$

For a continuous dividend yield,

$$q_\delta = \frac{e^{(r-\delta)h} - d}{u-d}, \qquad a = e^{-\delta h}\frac{v_U - v_D}{s(u-d)},$$

while dollar claim values continue to discount by R.

For a general finite-state market,

$$\begin{aligned} \text{no-arbitrage} &\iff \exists \psi > 0 :\ s = \psi \mathsf{A}, \\ \text{attainability of } X &\iff X \in \operatorname{im} \mathsf{A}, \\ \text{completeness, given no-arbitrage} &\iff \operatorname{rank}(\mathsf{A}) = M \\ &\iff \text{unique positive state-price vector}, \\ \mathcal{I}(X) &= \{\psi X : \psi > 0,\ \psi \mathsf{A} = s\}. \end{aligned}$$

With a positive numeraire, $q_j^N = \psi_j N_1(\omega_j)/N_0$ and $x_0 = N_0\mathbb{E}^{\mathbb{Q}^N}[X/N_1]$.

Chapter 3

Basic Stochastic Calculus

Preface

My notes assume that you are familiar with the basic objects of measure theory. I took ACTSC 446 with STAT 433 so the concepts were introduced more rigorously, along with their extensions to martingale theory in more depth in that course. For more rigorous treatment of σ-algebras, Borel sets, Hilbert spaces, Conditional Expectation as an orthogonal projection, and Doob martingales, please refer to my notes on *STAT 433*.

3.0.1 Brownian Motions and its Conditional Structure

Brownian Definitions relative to a Filtration

Definition 3.1 (Standard Brownian motion)**.** A *standard one-dimensional Brownian motion* relative to filtration $(\mathcal{F}_t)$ is an adapted process W such that $W_0 = 0$ almost surely, with paths that are almost surely continuous, and for every $s < t$,

$$W_t - W_s \sim \mathcal{N}(0, t-s), \qquad W_t - W_s \perp \mathcal{F}_s.$$

Hence, disjoint increments are independent as each successive increment is independent of the information containing all earlier increments. Conversely, a continuous process with independent *Gaussian* increments is Brownian relative to its natural filtration and its usual augmentation. We assume a space carrying such a process exists.

Brownian increments, on the contrary to typical Gaussian notation $\mathcal{N}(m, v)$, have standard deviation $\sqrt{t-s}$, not $t-s$. Hence, this square-root scaling is the source of the second-order term in stochastic calculus. We make note of important characteristics of Brownian increments next.

Moments, covariance, and conditional laws

Proposition 3.1 (Brownian moments and transition law)**.** *For $0 \leq s \leq t$,*

(i) $\mathbb{E}W_t = 0$ *and* $\mathbb{E}W_t^2 = t$,

(ii) $\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s$ *and* $Var(W_t \mid \mathcal{F}_s) = t - s$,

(iii) $Cov(W_s, W_t) = \min(s, t)$.

More precisely, for bounded Borel σ-algebra g and $s < t$,

$$\mathbb{E}[g(W_t) \mid \mathcal{F}_s] = \int_{\mathbb{R}} g(W_s + z) \frac{e^{-z^2/(2(t-s))}}{\sqrt{2\pi(t-s)}} dz.$$

Proof. Let $Z = W_t - W_s$. Then Z is clearly a centered Gaussian of variance $t - s$ independent of the filtration indexed $\mathcal{F}_s$. The decomposition, hence, $W_t = W_s + Z$ gives,

$$\begin{aligned}
\mathbb{E}[W_t \mid \mathcal{F}_s] &= W_s + \mathbb{E}[Z \mid \mathcal{F}_s] = W_s, \\
\mathbb{E}[W_t^2 \mid \mathcal{F}_s] &= W_s^2 + 2W_s\mathbb{E}[Z \mid \mathcal{F}_s] + \mathbb{E}[Z^2 \mid \mathcal{F}_s] \\
&= W_s^2 + t - s, \\
\mathrm{Var}(W_t \mid \mathcal{F}_s) &= W_s^2 + t - s - W_s^2 = t - s.
\end{aligned}$$

Taking $s = 0$ yields the first two moments of the Brownian increments. For covariance,

$$\begin{aligned}
\mathrm{Cov}(W_s, W_t) &= \mathbb{E}[W_s(W_s + Z)] \\
&= \mathbb{E}W_s^2 + \mathbb{E}[W_s\mathbb{E}[Z \mid \mathcal{F}_s]] \\
&= s.
\end{aligned}$$

By symmetry, we obtain the minimum formulation. Conditional-laws are obtained via independence, first for product functions of W_s and Z then for bounded Borel functions by a monotone-class argument. Finally, $\mathbb{E}[W_t] \leq \sqrt{t}$ so the conditional-mean identity indeed makes W a martingale.

Hence we have proved the properties.

□

The transition formulation is simply the *Markov property* where conditional on the present value, earlier observations do not alter the distribution of the future value. A joint vector $(W_{t_1}, \ldots, W_{t_n})$ is *Gaussian* because it is a linear transformation of independent *Gaussian* increments, hence its covariance matrix has entries $\min(t_i, t_j)$.

Proposition 3.2 (Two useful Brownian martingales)**.** *The processes $W_t^2 - t$ and,*

$$Z_t(\theta) = \exp\left(\theta W_t - \frac{1}{2}\theta^2 t\right), \qquad \theta \in \mathbb{R}$$

are martingales.

Proof. The conditional second-moment identity gives,

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 + (t - s) - t = W_s^2 - s.$$

For a centered *Gaussian* G of variance v, completing the square in its density yields,

$$\begin{aligned}\mathbb{E}[e^{\theta G}] &= \frac{1}{\sqrt{2\pi v}} \int_{\mathbb{R}} \exp\left(\theta z - \frac{z^2}{2v}\right) dz \\ &= e^{\theta^2 v/2} \frac{1}{\sqrt{2\pi v}} \int_{\mathbb{R}} \exp\left(-\frac{(z-\theta v)}{2v}\right) dz \\ &= e^{\theta^2 v/2}.\end{aligned}$$

Hence, $\mathbb{E}Z_t(\theta) = 1$ and, by independence of future increments,

$$\begin{aligned}\mathbb{E}[Z_t(\theta) \mid \mathcal{F}_s] &= Z_s(\theta)\mathbb{E}\left[e^{\theta(W_t - W_s) - \theta^2(t-s)/2}\middle|\mathcal{F}_s\right] \\ &= Z_s(\theta).\end{aligned}$$

□

3.0.2 Indifferentiability of Brownian Paths

For a fixed t and $h > 0$,

$$\frac{W_{t+h} - W_t}{h} \sim \mathcal{N}(0, 1/h).$$

As we let $h \to 0$, the variance diverges. However, this claim alone does not prove simultaneous nondifferentiability at every timestep. The following argument supplies that missing step.

Theorem 3.1 (Nowhere Differentiable)**.** *Almost surely, a Brownian path has no finite derivative at any inferior time of a compact interval. It also has no finite one-sided derivative at a specified endpoint. Hence, it is nondifferentiable everywhere, almost surely.*

Proof. Suppose we are working on an interval $[0, T]$ for $T > 0$ and let $h_n = T2^{-n}$. Suppose that a path has finite derivative at some $t < T$. Then there exists $M, m \geq 1 \in \mathbb{R}$ such that,

$$|W_u - W_t| \leq M|u - t| \qquad \text{whenever } |u - t| < \frac{1}{m}.$$

For all sufficiently fine grids (or partitions), let k satisfy the inequality,

$$kh_n \leq t < (k+1)h_n.$$

Then, the three adjacent grid increments with endpoints,

$$(k+1)h_n, \quad (k+2)h_n, \quad (k+3)h_n, \quad (k+4)h_n$$

lies to the right of t and inside the indicated neighbourhood. Each is bounded in absolute value by $8Mh_n$, by the triangle inequality through W_t. For any specified block, these increments are independent of $\mathcal{N}(0, h_n)$ variables. Since a standard normal density is at most $\frac{1}{\sqrt{2\pi}}$, □

$$\mathbb{P}(|\mathcal{N}(0, h_n)| \leq 8Mh_n) \leq CM\sqrt{h_n}$$
$$\mathbb{P}(\text{some block has all three bounds}) \leq 2^n (CM\sqrt{h_n})^3 = C_M\sqrt{h_n}.$$

The last bounds are summable in n.